The CRM Signature of Berkovich Motives: How Scholze's Independence of ℓ Confirms the Archimedean Obstruction Thesis

Paper 101 of the Constructive Reverse Mathematics Series

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Abstract

We perform a constructive reverse mathematics (CRM) audit of Scholze's Berkovich motivic proof of independence of ℓ for local Langlands parameters (2024–2026). The classical approach to independence of ℓ requires the isomorphism $\iota : \overline{\mathbb{Q}_\ell} \cong \mathbb{C}$, routing the comparison through the Archimedean place at cost CLASS. Scholze's strategy—replacing ℓ-adic sheaves with motivic sheaves that are natively independent of ℓ—achieves a strictly lower CRM signature. We trace the logical dependencies of the complete proof and establish:

(i) The CRM signature of the Berkovich motivic proof is WKL (after excision of a parasitic WLPO), strictly below the CLASS signature of all previously known approaches. The literal syntax of the general Berkovich formalism requires BISH + WKL + WLPO (since WKL and WLPO are incomparable over BISH), but the WLPO is a dispensable artifact of $\mathbb{R}$-valued norms in a context where all relevant value groups are contained in $\mathbb{Q}$ and hence algebraically decidable.

(ii) The proof is logically independent of the ℓ-adic Fargues–Scholze construction: Scholze rebuilds the geometric Satake equivalence, the spectral action, and the Langlands parameterization natively within the motivic 6-functor formalism, extracting a universal motivic parameter φ_{mot} before any realization functor is applied.

(iii) The WKL cost is structurally essential, arising from the non-Archimedean compactness (inverse limits of finite rings) required to construct perfectoid spaces and the Fargues–Fontaine curve. The WLPO is parasitic, arising from the formalism's use of $\mathbb{R}$-valued Banach norms in a context where all relevant value groups are $\mathbb{Q}$-valued (hence decidable).

(iv) The Archimedean Obstruction Thesis (Paper 98) correctly predicts the signature: any proof confined to non-Archimedean realizations and comparisons has signature at most WKL, and the motivic "stay at the bedrock" strategy achieves this bound.

This represents the first CRM analysis of frontier arithmetic geometry produced independently of the CRM programme, and establishes that two programmes—constructive reverse mathematics and Berkovich motivic geometry—converged on the same structural diagnosis of the Archimedean obstruction without mutual awareness or communication.

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1 Introduction

1.1 The Independence of ℓ Problem

Let E be a p-adic local field and G a reductive group over E. The Fargues–Scholze geometrization of the local Langlands correspondence [3] constructs, for each irreducible smooth representation π of $G(E)$, a semisimple L-parameter

$$\varphi_{\pi,\ell} : W_E \longrightarrow {}^L G(\overline{\mathbb{Q}_\ell})$$

using the geometry of G-bundles on the Fargues–Fontaine curve and ℓ-adic étale cohomology, for an auxiliary prime $\ell \neq p$.

The construction depends, a priori, on the choice of ℓ. The *independence of ℓ conjecture* [3, Conjecture I.9.5] asserts that the association $\pi \mapsto \varphi_{\pi,\ell}$ is independent of ℓ (and of the isomorphism $\iota : \overline{\mathbb{Q}_\ell} \cong \mathbb{C}$ used to normalize it).

1.2 The Classical Approach: Why It Costs CLASS

All previously known strategies for proving independence of ℓ require *globalizing*: embedding the local situation into a global one over a number field, and then using the global Arthur–Selberg trace formula to establish that the Galois representations attached to automorphic forms are independent of ℓ. This strategy incurs CLASS costs at two points:

(a) **The trace formula.** The spectral decomposition of $L^2(G(\mathbb{Q})\backslash G(\mathbb{A}_\mathbb{Q}))$ requires harmonic analysis on the Archimedean factor $G(\mathbb{R})$: Plancherel measure, Harish-Chandra characters, spectral projections. This is irreducibly CLASS.

(b) **The comparison isomorphism.** Comparing ℓ_1-adic and ℓ_2-adic representations requires a common coefficient field, classically obtained via the isomorphism $\iota : \overline{\mathbb{Q}_\ell} \cong \mathbb{C}$. Constructing this isomorphism requires the Axiom of Choice for continuum-sized sets (CLASS).

The Archimedean Obstruction Thesis (Paper 98, [4]) predicts this: any proof path that passes through the Archimedean place—whether via the trace formula, period integrals, or the embedding into $\mathbb{C}$—has CRM signature $\geq$ CLASS.

1.3 Scholze's Bypass: Motivic Descent

In two remarkable papers [1, 2], Scholze resolves the independence of ℓ conjecture by an entirely different strategy. Instead of comparing ℓ-adic constructions *a posteriori*, he builds the Langlands parameterization *natively* in a universal motivic coefficient system—Berkovich motives—that is independent of ℓ by construction. The ℓ-adic parameters are then recovered as specializations of a single motivic parameter φ_{mot}.

This paper performs a CRM audit of this proof, tracing the logical resources required at each step and establishing the overall CRM signature.

1.4 Motivic Descent as a Fourth Mode

Paper 98 identified three modes of *de-omniscientising descent*—strategies for reducing the CRM cost of a proof:

(i) **Algebraic bypass**: replace an analytic computation with an algebraic one (e.g., Taylor–Wiles patching replacing the Petersson inner product).

(ii) **Locus restriction**: restrict to a special sublocus where algebraic shortcuts exist (e.g., the palindromic locus for Hodge, $\rho = 20$ K3 surfaces for Kuga–Satake).

(iii) **Domain transfer**: move to a domain lacking the Archimedean place (e.g., function fields $\mathbb{F}_q(C)$).

Scholze's strategy constitutes a fourth mode:

(iv) **Motivic descent**: construct the mathematical object at the motivic level (the "bedrock"), before any realization functor is applied. The realizations become derived consequences rather than primary constructions. Since the motive is BISH (Paper 98, Proposition 3.1) and no realization or comparison isomorphism is invoked, the CRM signature is determined entirely by the structural cost of the motivic construction itself.

The first three modes operate at the level of realizations and attempt to avoid specific CLASS bottlenecks. Motivic descent operates below the realizations entirely.

2 The CRM Framework

We recall the essential definitions from Paper 98 [4]. The CRM hierarchy is a partial order of logical principles:

$$\mathsf{BISH} \subsetneq \mathsf{LLPO} \subsetneq \mathsf{WLPO} \subsetneq \mathsf{LPO} \subsetneq \mathsf{CLASS}$$

with WKL (Weak König's Lemma) incomparable with WLPO over BISH.

Definition 2.1 (CRM Signature of a Proof Path)**.** A proof path P using realizations $\{R_i\}$ and comparison isomorphisms $\{C_j\}$ has signature:

$$\sigma(P) = \max\big(\sigma(\text{motive}),\ \max_i \sigma(R_i),\ \max_j \sigma(C_j)\big)$$

Theorem 2.2 (Archimedean Obstruction, Paper 98)**.** *If P uses only non-Archimedean realizations and comparisons, then $\sigma(P) = \mathsf{BISH}$. If P uses any Archimedean realization or comparison, then $\sigma(P) \geq \mathsf{CLASS}$.*

The key prediction: a proof that *stays motivic*—never applying a realization functor—has signature $\sigma(P) = \sigma(\text{motive}) = \mathsf{BISH}$, modulo the structural cost of the motivic construction itself.

3 CRM Audit of the Berkovich Motivic Proof

We now trace the logical dependencies of Scholze's proof through its major components.

3.1 The Berkovich Motivic Category $D_{\mathrm{mot}}(X)$

CRM Audit 3.1 (Berkovich Motivic Category)**.** $\sigma(D_{\mathrm{mot}}(X)) = \mathsf{WKL}$ (with parasitic WLPO).

The construction of $D_{\mathrm{mot}}(X)$ proceeds through several layers: Banach rings, the arc-topology, finitary arc-sheaves, ball-invariant sheaves, and the Tate twist inversion.

The WKL content is structurally essential. The arc-topology on Banach rings requires checking covers by valuation rings. For affinoid algebras over local fields (the only case required for local Langlands), the existence of sufficiently many valuation rings to populate the Berkovich spectrum $\mathcal{M}(A)$ reduces to extracting paths from profinite trees—precisely WKL. In the general setting, one would invoke Zorn's Lemma or the Boolean Prime Ideal theorem (CLASS), but the local Langlands application restricts to p-adic affinoid algebras where WKL suffices.

The parasitic WLPO arises because Banach rings carry norms valued in $\mathbb{R}_{\geq 0}$, and testing inequalities of arbitrary real numbers requires WLPO in general. However, the value groups that actually arise in the proof—including those of perfectoid fields such as $\mathbb{C}_p$ and its tilt $\mathbb{C}_p^\flat$—are

contained in $\mathbb{Q}$ (specifically, in $p^{\mathbb{Q}} \cup \{0\}$). While these value groups are dense in $\mathbb{R}_{>0}$ (perfectoid fields are not discretely valued by definition), order and equality in $\mathbb{Q}$ are algebraically decidable (BISH). The dependence on the full real continuum $\mathbb{R}_{\geq 0}$ is an artifact of the general Berkovich formalism, not a structural requirement of the proof.

Remark 3.2 (The Norm Artifact)**.** This parasitic WLPO is analogous to the false positives detected by the CRM scanner in Paper 99: a surface-syntax feature (real-valued norms) that does not reflect the logical content of the proof ($\mathbb{Q}$-valued norms with decidable ordering). A Lean 4 formalization using an adic foundational framework restricted to $\mathbb{Q}$-valued norms would natively bypass WLPO while faithfully representing the dense value groups of perfectoid fields.

3.2 Recovery of Voevodsky's Algebraic Motives

CRM Audit 3.3 (Voevodsky Recovery)**.** $\sigma(D_{\mathrm{mot}}(k) \cong DM(k)) = \mathsf{BISH}$.

For a discrete field k, the Berkovich spectrum collapses (the norm is trivial), and the equivalence $D_{\mathrm{mot}}(k) \cong DM(k)$ with Voevodsky's étale motives is established via $\mathbb{A}^1$-homotopy invariance, Nisnevich descent, and algebraic correspondences. This is a substantial categorical theorem, but every ingredient is purely algebraic: no limits, no completions, no analysis.

3.3 The Geometric Satake Equivalence (Motivic Version)

CRM Audit 3.4 (Motivic Geometric Satake)**.** $\sigma(\text{motivic Satake}) = \mathsf{BISH}$.

This is a critical potential leak point. The *classical* geometric Satake equivalence uses perverse sheaves and intersection cohomology on the complex affine Grassmannian, routing the proof through the Archimedean topology (CLASS). If the motivic version depended on the classical version, CLASS would enter through the back door.

It does not. The motivic geometric Satake equivalence, as developed by Richarz–Scholbach [5] and Cass–van den Hove–Scholbach [6], is rebuilt natively within the motivic category. The affine Grassmannian is an ind-algebraic scheme; the equivalence is constructed using algebraic weight structures, Wakimoto filtrations, and Mirković–Vilonen cycles (algebraic subvarieties). The proof explicitly avoids the ℓ-adic and Betti versions and bypasses the Riemann Existence Theorem (CLASS). The logical resources are pure algebra: BISH.

3.4 The Fargues–Fontaine Curve and Bun_G

CRM Audit 3.5 (Fargues–Fontaine and Bun_G)**.** $\sigma(\text{Fargues–Fontaine}) = \mathsf{WKL}$.

The Fargues–Fontaine curve and the stack Bun_G of G-bundles are fundamentally p-adic, non-Archimedean objects. The curve is constructed from Witt vectors and perfectoid tilts. Its geometry is controlled by Newton polygons and Dieudonné theory—algebraic invariants of p-divisible groups.

The WKL cost is essential: perfectoid tilting is defined via the inverse limit $R^\flat = \varprojlim_{x \mapsto x^p} R/p$ of discrete finite rings, and perfectoid spaces arise geometrically as inverse limits of towers of finite-type rigid spaces. Extracting compatible systems from these profinite inverse limits is exactly WKL. The stratification of Bun_G (into Harder–Narasimhan strata) is geometric and algebraic. No Archimedean input is required.

Remark 3.6 (Spectral Spaces and the WKL Lock)**.** The underlying topological spaces of perfectoid spaces and Huber's adic spaces are *spectral spaces* in the sense of Hochster [14]. In classical reverse mathematics, the compactness of spectral spaces via Stone duality is equivalent to the Boolean Prime Ideal Theorem (BPI) in general. However, for countably generated p-adic algebras—the only case required for local Langlands—the compactness reduces to the compactness of profinite trees, which is exactly WKL. This securely locks the cost at WKL and provides a precise mathematical anchor for the classification.

3.5 The Spectral Action and Excursion Operators

CRM Audit 3.7 (Spectral Action)**.** σ(spectral action) = BISH.

The terminology "spectral action" is potentially misleading from a CRM perspective. In the classical Langlands programme, "spectral" refers to the L^2 spectral decomposition of automorphic forms on the adèlic quotient $G(\mathbb{Q})\backslash G(\mathbb{A})$—the universal CLASS bottleneck (Paper 98, §6.2).

In Scholze's construction, "spectral" refers to the algebraic spectrum (Spec) of the Bernstein center, a commutative ring. Excursion operators are categorical endomorphisms of the identity functor on $D_{\mathrm{mot}}(\mathrm{Bun}_G)$, constructed from correspondences on the Hecke stack. The construction of a commutative ring of endomorphisms in a categorical setting is pure abstract algebra. No L^2 spaces, no harmonic analysis on $G(\mathbb{R})$, no Plancherel measure: BISH.

3.6 Descent from Motivic to ℓ-adic

CRM Audit 3.8 (ℓ-adic Realization)**.** σ(ℓ-adic realization) = WKL.

After constructing the motivic parameter φ_{mot}, the ℓ-adic parameter is recovered by applying the étale realization functor $R_\ell : D_{\mathrm{mot}} \to D_{\text{ét}}(\cdot, \mathbb{Q}_\ell)$. Computing étale cohomology with finite coefficients $\mathbb{Z}/\ell^n\mathbb{Z}$ is finite algebraic geometry (BISH). Passing to $\mathbb{Z}_\ell$-coefficients requires the inverse limit $\varprojlim_n H^*(\cdot, \mathbb{Z}/\ell^n\mathbb{Z})$, which invokes the compactness of profinite systems (WKL).

Critically, both domain and target of the realization functor are non-Archimedean. No CLASS comparison isomorphism is invoked. The ℓ-adic parameter is a *specialization* of the motivic parameter, not a comparison between realizations.

3.7 Archimedean Content: Absence

CRM Audit 3.9 (Archimedean Input)**.** None.

Scholze's architecture operates entirely over non-Archimedean local fields and the residue field $\mathbb{F}_q$. There are no period integrals, no Hodge decompositions, no $\mathbb{R}$-manifolds, no analytic continuations of L-functions, no classical trace formulas, and no embeddings $\iota : \overline{\mathbb{Q}_\ell} \cong \mathbb{C}$. The proof never touches the Archimedean place.

4 Overall CRM Signature

Theorem 4.1 (CRM Signature of Berkovich Motivic Independence of ℓ)**.** *The CRM signature of Scholze's proof of the independence of ℓ conjecture [1, 2] is WKL, after excision of a parasitic WLPO.*

Proof. By the signature formula $\sigma(P) = \max(\sigma(\text{inputs}))$, we take the join over all components. We must proceed in two steps, distinguishing the literal syntax of the general formalism from the essential structure of the specific proof.

Step 1: Literal syntactic signature. The general Berkovich formalism operates on Banach rings with $\mathbb{R}_{\geq 0}$-valued norms. Testing inequalities of arbitrary real numbers requires WLPO. The profinite inverse limits require WKL. Since WKL and WLPO are *incomparable* over BISH, their join in the lattice of logical principles is strictly above both: the literal syntax requires BISH + WKL + WLPO.

Step 2: Essential structural signature. However, the independence of ℓ proof operates exclusively over p-adic local fields and their perfectoid extensions. While perfectoid fields such as $\mathbb{C}_p$ possess dense value groups (not discrete), these groups are contained in $\mathbb{Q}$ (specifically $p^{\mathbb{Q}} \cup \{0\}$), where order and equality are algebraically decidable (BISH). The dependence on the full real continuum $\mathbb{R}_{\geq 0}$ is a dispensable artifact of the general Berkovich formalism, not a

structural requirement of the proof. By reformulating the foundations using $\mathbb{Q}$-valued norms—for instance, via a variant of Huber's adic spaces with $\mathbb{Q}$-valued ordered value groups—the parasitic WLPO is formally excised. The structurally essential components are:

Component	**CRM Level**	**Justification**
Berkovich motivic category $D_{\mathrm{mot}}(X)$	WKL	Profinite inverse limits (arc-topology)
Voevodsky recovery	BISH	$\mathbb{A}^1$-homotopy, Nisnevich descent
Motivic geometric Satake	BISH	Algebraic MV cycles, weight structures
Fargues–Fontaine curve, Bun_G	WKL	Perfectoid inverse limits
Spectral action / excursion	BISH	Categorical endomorphisms, Spec of ring
ℓ-adic realization	WKL	$\varprojlim \mathbb{Z}/\ell^n\mathbb{Z}$
Archimedean input	—	None

Table 1: CRM audit of Scholze's Berkovich motivic proof (essential structure).

After excision of the parasitic WLPO, the maximum of the structurally essential components is exactly WKL. □

Remark 4.2 (Historical Comparison)**.** The CRM signature of the *classical* approach to independence of ℓ (globalization + trace formula + ι-comparison) is CLASS. Scholze's motivic descent achieves a strict reduction: CLASS $\to$ WKL. This is the same type of logical excision documented for the Taylor–Wiles fix of Fermat's Last Theorem (CLASS $\to$ WKL, Paper 98, Theorem 8.1), but at a far greater level of mathematical sophistication.

5 Logical Independence

A critical question for the CRM audit is whether Scholze's motivic proof is *logically independent* of the ℓ-adic Fargues–Scholze construction, or whether it uses the ℓ-adic results as input and "motivifies" them.

Theorem 5.1 (Logical Independence)**.** *The Berkovich motivic proof of independence of ℓ is logically independent of the ℓ-adic Fargues–Scholze construction. The motivic parameter φ_{mot} is constructed before any realization functor is applied.*

Argument. Scholze explicitly reconstructs the entire architecture—the geometry of Bun_G, the geometric Satake equivalence, and the categorical spectral action—natively within the motivic 6-functor formalism $D_{\mathrm{mot}}(\mathrm{Bun}_G)$. The universal motivic Langlands parameter φ_{mot} is extracted from the spectral action on D_{mot} before any realization functor is applied. The ℓ-adic parameter is then recovered by:

$$\varphi_{\pi,\ell} = R_\ell(\varphi_{\mathrm{mot}})$$

where R_ℓ is the étale realization functor. Because the proof never compares R_{ℓ_1} and R_{ℓ_2} horizontally (i.e., never invokes a comparison isomorphism between different ℓ-adic cohomology theories), no CLASS comparison cost is incurred. Independence of ℓ follows formally: all ℓ-adic specializations derive from the single motivic object φ_{mot}.

If, instead, the motivic version had used the ℓ-adic Fargues–Scholze as a black box and then shown independence by comparing realizations, the proof would require $\iota : \overline{\mathbb{Q}_\ell} \cong \mathbb{C}$ or an analogous comparison, introducing CLASS. The natively motivic construction avoids this entirely. □

6 Condensed Mathematics: An Orthogonal Elimination

It is instructive to compare Scholze's two major foundational programmes—condensed mathematics and Berkovich motives—from the CRM perspective.

Proposition 6.1 (Orthogonality of Condensed and Motivic Eliminations)**.** *Condensed mathematics and Berkovich motives eliminate logically independent sources of* CLASS *content:*

(i) ***Condensed mathematics*** *eliminates* CLASS *content arising from* topological foundations*: Baire category, Tychonoff compactness, Zorn's lemma for locally convex spaces. It replaces topological spaces with condensed sets (sheaves on profinite sets), reducing the cost of continuous functional analysis from* CLASS *to* WKL*.*

(ii) ***Berkovich motives*** *eliminate* CLASS *content arising from* cohomological comparisons*: the isomorphism $\iota : \overline{\mathbb{Q}_\ell} \cong \mathbb{C}$, comparison between ℓ-adic theories, period integrals. They achieve this by working in a universal motivic coefficient system that is independent of ℓ by construction.*

In the language of Paper 98: condensed mathematics fixes the *target categories* of the realization functors (making them better-behaved abelian categories), while Berkovich motives eliminate the *realization step itself* (making it unnecessary by working at the motivic level). One fixes the topology of the base spaces. The other fixes the coefficients of the cohomology. Both are needed for a complete elimination of CLASS from the non-Archimedean foundations; but for the specific problem of independence of ℓ, it was the cohomological comparison (not the topological foundation) that constituted the obstruction. This explains why condensed mathematics alone—despite its profound restructuring of functional analysis—could not resolve independence of ℓ.

7 Convergence of Two Programmes

The most striking feature of this analysis is the *convergence* of two independent programmes on the same structural diagnosis.

The CRM programme (Papers 50–100) developed the Archimedean Obstruction Thesis: $\mathbb{R}$ is the unique source of non-constructive content in arithmetic geometry. The thesis predicts that any proof confined to non-Archimedean realizations and comparisons has CRM signature at most WKL, and that the optimal strategy for reducing logical cost is to avoid the Archimedean place.

Scholze's programme (perfectoid spaces, diamonds, condensed mathematics, Berkovich motives) developed the geometric infrastructure for the local Langlands correspondence. The motivic strategy was adopted to resolve independence of ℓ—a problem internal to the Langlands programme, unrelated to constructive mathematics.

The two programmes are *unaware of each other*. Scholze does not use CRM vocabulary, does not reference constructive mathematics, and did not set out to reduce logical cost. The CRM programme does not use ∞-categories, diamonds, or 6-functor formalisms.

Yet both programmes identify the same obstruction (the Archimedean place), predict the same consequence (passing through $\mathbb{R}$ is the source of the difficulty), and arrive at the same bypass strategy (stay in the non-Archimedean world). The CRM programme expresses this as: $\mathbb{R}$ introduces CLASS through period integrals and spectral decomposition. Scholze's programme expresses this as: $\mathbb{R}$ introduces ℓ-dependence through the isomorphism $\iota : \overline{\mathbb{Q}_\ell} \cong \mathbb{C}$. Same obstruction, different descriptions, different reasons for caring, same structural bypass.

This convergence provides evidence that the Archimedean obstruction is an *objective structural feature of the mathematical landscape*, not a framework-dependent artifact of either the CRM classification or the motivic formalism. Two independent diagnostic systems, applied to

overlapping but distinct bodies of mathematics, identify the same singular point ($\mathbb{R}$) as the source of the difficulty and the same strategy (avoidance) as the resolution.

8 Predictions

If the "stay motivic" strategy successfully avoids realization costs for independence of ℓ, the Archimedean Obstruction Thesis generates predictions for other results currently proved using ℓ-adic methods.

Prediction 8.1 (Weight-Monodromy Conjecture (Known Cases)). $\sigma = \mathsf{WKL}$. The conjecture governs the action of the Weil group on cohomology. If formulated natively in D_{mot} via motivic weight filtrations, the proof stays non-Archimedean. The residual obstruction is the p-adic inverse limits and perfectoid geometry (WKL), cleanly bypassing Betti–étale comparisons over $\mathbb{C}$.

Prediction 8.2 (Langlands Correspondence for $\mathrm{GL}_n/\mathbb{F}_q(C)$). $\sigma = \mathsf{BISH}$. Lafforgue's proof is already BISH (Paper 98, Theorem 10.1): the adèle ring over $\mathbb{F}_q(C)$ has no Archimedean factor, and the proof uses moduli of shtukas (algebraic) rather than the trace formula. A purely motivic proof via D_{mot} over $\mathbb{F}_q$ would provide a structurally cleaner, ℓ-independent BISH proof. Since function field curves are discrete/algebraic, even the WKL perfectoid completion cost is avoided.

Prediction 8.3 (Ramanujan–Petersson for $\mathrm{GL}_2/\mathbb{F}_q(C)$). $\sigma = \mathsf{BISH}$. Deligne's proof uses the geometry of moduli of shtukas and the Grothendieck–Lefschetz trace formula over finite fields—purely algebraic intersection theory, avoiding the $L^2(G(\mathbb{R}))$ classical trace formula. This lifts to the motivic level at BISH.

Prediction 8.4 (Tate Conjecture for Abelian Varieties over $\mathbb{F}_q$). $\sigma = \mathsf{BISH}$. Tate's original 1966 proof relies on the Riemann hypothesis for abelian varieties (Weil conjectures) and the finiteness of isogeny classes of a given dimension over $\mathbb{F}_q$. A natively motivic proof would execute this linear algebra entirely within $DM(\mathbb{F}_q)$, remaining at BISH.

Prediction 8.5 (Independence of ℓ for Shimura Varieties). $\sigma = \mathsf{CLASS}$ (irreducible obstruction). The "stay motivic" strategy fails here. Shimura varieties are global objects defined over number fields; they inherently possess Archimedean places. Proving that the motivic Galois representation matches the Langlands parameter of an automorphic form requires Matsushima's formula and the global Arthur–Selberg trace formula. The trace formula irreducibly requires the L^2 spectral decomposition of $G(\mathbb{R})$. The motivic framework cannot bypass the global Archimedean geometry of the adèle ring $\mathbb{A}_{\mathbb{Q}}$. The CLASS obstruction is structural and permanent.

Remark 8.6 (The Shimura Boundary). Prediction 8.5 identifies the *limit* of motivic descent: it works for local problems (local Langlands, where the Archimedean place is absent) but not for global problems (Shimura varieties, where the Archimedean place is intrinsic to the geometry). The Archimedean Obstruction Thesis correctly predicts both where the strategy succeeds and where it structurally cannot succeed. This predictive power—identifying limits as well as opportunities—is the strongest evidence for the thesis's validity.

9 Scanner Validation

As independent confirmation of the architectural CRM audit (§3), we applied the CRMlint LaTeX scanner (Paper 76; DOI: 10.5281/zenodo.18779362) to the TEX sources of [1] and [2], obtained from the arXiv e-print server. The scanner employs ∼91 keyword patterns and ∼120 citation patterns against a CRM-classified dictionary derived from Papers 50–97, augmented with 11 Berkovich-specific entries (perfectoid, arc-topology, Berkovich spectrum, condensed, solid module, adic space, motivic sheaf, six-functor, Tate twist).

9.1 Quantitative Results

The scanner produced 214 alerts across both papers.[1]

CRM Level	Berkovich Motives	Motivic Geom.	Total
CLASS	0	0	0
LPO	7	1	8
WLPO	41	27	68
WKL	51	25	76
BISH	42	20	62
Total	141	73	214

Table 2: Scanner alert counts by CRM level. The CLASS row is uniformly zero.

Each alert was manually classified into four categories: **A** (motivational—classical approach described in introduction), **B** (parasitic—Banach ring formalism carrying algebraic content), **BISH** (algebraic core—genuine constructive content), and **C** (structural WKL—irreducible profinite/inverse limit constructions).

Category	Berkovich	Motivic Geom.	Total	%
A (motivational)	14	12	26	12.1%
B (parasitic WLPO/LPO)	46	19	65	30.4%
BISH (algebraic core)	42	20	62	29.0%
C (structural WKL)	39	22	61	28.5%
Total	141	73	214	100%

Table 3: Alert classification. Category B (parasitic) dominates the non-algebraic alerts; Category C (structural) is exclusively WKL.

9.2 Key Findings

(i) Zero CLASS alerts. Across 93 combined pages, 34 theorems, 91 lemmas, 109 proofs, and 87 bibliographic references, the scanner found no CLASS-level alert. No invocation of the Axiom of Choice, Zorn's Lemma, well-ordering, harmonic analysis on $G(\mathbb{R})$, Plancherel measure, or Harish-Chandra characters was detected in either paper. This independently confirms that Scholze's motivic programme operates entirely below CLASS.

(ii) Parasitic dominance. Category B alerts (65 of 214, or 30.4%) arise from the Banach ring formalism—normed completions, seminorm filtrations, completed normed groups—propagating through the proof. All 8 LPO alerts and 57 of the 68 WLPO alerts are parasitic. These alerts correctly detect a surface anomaly: real-analytic terminology in a proof whose essential structure is non-Archimedean. In a reformulation using Huber's adic spaces restricted to decidable $\mathbb{Q}$-valued groups, all Category B alerts would vanish.

(iii) Structural WKL concentration. The 61 Category C alerts cluster in the expected locations: perfectoid tilting (arc-topology sections), inverse limits of finite rings (nonarchimedean unit disc), condensed mathematics citations (Lectures on Analytic Geometry), and profinite descent (Fargues–Fontaine infrastructure). This matches the architectural audit (§3) exactly.

[1] Of these, 10 were identified as anti-marker suppressions (7 in [1], 3 in [2])—motivational or contextual citations that match scanner patterns but do not reflect proof-level logical content. These 10 are included in all quantitative tables for transparency and reproducibility.

(iv) Algebraic core verification. The 62 BISH alerts—motivic sheaves, Tate twists, six-functor formalism, Lafforgue citations, Fontaine-Laffaille theory—confirm that the core mathematical content is genuinely constructive, consistent with the CRM audit's assessment that the motivic geometric Satake equivalence, the spectral action, and excursion operators are all BISH.

9.3 Cross-Validation

The scanner output was cross-validated against the five structural claims of the architectural audit (§3):

(a) Scanner finds no CLASS markers in geometric Satake sections ✓ (confirms Audit 3.4: BISH)

(b) Scanner finds WKL markers in Fargues–Fontaine/perfectoid sections ✓ (confirms Audit 3.5: WKL)

(c) Scanner finds no L^2/harmonic analysis/Plancherel markers in spectral action sections ✓ (confirms Audit 3.7: BISH)

(d) Scanner finds WLPO markers in Banach ring foundation sections ✓ (confirms Audit 3.1: parasitic WLPO)

(e) Scanner finds CLASS markers only in motivation ✓ (confirms Audit 3.9: no Archimedean input)

All five checks pass. The full classified alert list (214 entries) is provided in the supplementary appendix [18] for reproducibility.

10 Epistemological vs. Geometric Motives: Reconciling the Frameworks

Paper 98's motives and Scholze's motives operate at different levels of abstraction but reflect the same underlying mathematical reality. The distinction is illuminating: Grothendieck's motives are *epistemological*—they classify the logical content of cohomological constructions—while Scholze's Berkovich motives are *geometric*—they provide a computational 6-functor formalism on analytic spaces. The former are algebraic and discrete (hence natively BISH); the latter are analytic and topological (hence requiring WKL for the profinite completions). This distinction perfectly mirrors the logical jump from BISH to WKL.

Paper 98 classifies Grothendieck's category of Chow motives as BISH: the objects (algebraic correspondences) and morphisms (intersection theory modulo rational equivalence) are defined by finitary algebraic operations. This is a *logical* classification—it concerns what axioms are needed to define and manipulate the category.

Scholze's Berkovich motives are a *computational* construction: a 6-functor formalism on Banach rings that produces motivic sheaves specializing to ℓ-adic cohomology for any ℓ. His construction sits at WKL not because he checked its axiom dependencies, but because it operates entirely within the non-Archimedean world while requiring profinite inverse limits for the perfectoid foundations.

The relationship: Paper 98 explains *why* Scholze's strategy works. The signature formula $\sigma(P) = \max(\sigma(\text{motive}), \max_i \sigma(R_i), \max_j \sigma(C_j))$ predicts that a proof avoiding all realizations and comparisons has signature $\sigma(\text{motive}) = \mathsf{BISH}$. Scholze's construction achieves this (plus the structural WKL for profinite limits) by staying at the motivic level. The CRM classification provides the *logical* explanation for a *mathematical* phenomenon.

The decidability that Paper 98 identifies as BISH at the motivic level is the *same algebraic finiteness* that enables Scholze's construction to work without choosing ℓ. Algebraic operations

over $\mathbb{Q}$ and its non-Archimedean completions are finitary and do not require the continuum. Paper 98 expresses this as "these operations are BISH." Scholze expresses this as "these operations can be performed in a universal motivic category before choosing ℓ." Same fact, different languages.

11 Conclusion

The CRM audit of Scholze's Berkovich motivic proof of independence of ℓ establishes:

(1) The proof has CRM signature WKL, strictly below the CLASS signature of all previously known approaches.

(2) The reduction CLASS $\to$ WKL is achieved by motivic descent—a fourth mode of de-omniscientising descent not identified in Paper 98.

(3) The proof is logically independent of the ℓ-adic Fargues–Scholze construction.

(4) The Archimedean Obstruction Thesis correctly predicts the signature.

(5) The convergence of two independent programmes on the same structural diagnosis provides evidence that the Archimedean obstruction is an objective feature of the mathematical landscape.

The five predictions (§8) are testable: as additional results are proved motivically, their CRM signatures can be audited by the same methodology. The prediction that independence of ℓ for Shimura varieties remains irreducibly CLASS identifies the structural boundary of motivic descent.

Acknowledgments. The author is an interventional cardiologist, not a professional mathematician; all CRM classifications are heuristic, based on architectural analysis of the published proofs and validated by the CRMlint scanner. The author thanks Claude (Anthropic) for assistance with the analysis and preparation of this manuscript.

A Formalization Map: CRM Audit Components and Lean Feasibility

This appendix provides a structured map of the CRM audit components from §3, classifying each by CRM level, proof technique, and feasibility of Lean 4 formalization. The map is intended to guide formalization efforts and prioritize targets for the CRMlint Lean 4 backend (Paper 76, Generation 4).

A.1 Component-Level Classification

Table 4 lists each audited proof component with its CRM level, the key techniques involved, and a feasibility assessment for Lean 4 formalization. Feasibility ratings:

HIGH: The mathematical content is within reach of current Lean 4/Mathlib infrastructure. Finite algebraic verifications, ring/field arithmetic, decidable properties.

MEDIUM: Requires Mathlib libraries that exist but need adaptation (e.g., category theory, perfectoid foundations, inverse limits). Feasible with moderate effort.

LOW: Requires substantial new Mathlib development (e.g., motivic categories, 6-functor formalisms, derived ∞-categories). Multi-year project.

STRUCTURAL: The CRM classification itself (not the underlying mathematics) is formalizable: the axiom audit, logical dependency tracing, and signature computation can be encoded in Lean 4 as a metatheoretic framework.

Table 4: Formalization map for CRM audit of Berkovich motivic independence of ℓ. CRM levels and Lean 4 feasibility per component.

Component	**CRM**	**Lean**	**Key Techniques / Dependencies**	**Ref**
CRM Framework (metatheory)				
CRM hierarchy	BISH	HIGH	Finite partial order on 7 principles. `inductive CRMLevel`. Done in Papers 72–76.	§2
Signature formula $\sigma(P)$	BISH	HIGH	max over finite set. Lean 4 `Finset.sup`.	§2
Archimedean Obstruction Thm	BISH	HIGH	Metatheorem on `CRMLevel`. Paper 98 Lean bundle.	Thm 2.2
Berkovich Motivic Category $D_{\mathrm{mot}}(X)$ — Audit 3.1				
Banach ring foundations	WLPO*	MED	$\mathbb{R}_{\geq 0}$ norms, completions. Mathlib `SeminormedRing`. *Parasitic.	3.1
Arc-topology descent	WKL	LOW	Grothendieck topology on Ban_k. New infrastructure needed.	3.1
Profinite inverse limits	WKL	MED	$\varprojlim$ of finite rings. Perfectoid library partially covers.	3.1
Berkovich spectrum $\mathcal{M}(A)$	WKL	LOW	Bounded multiplicative seminorms. Not in Mathlib.	3.1
Voevodsky Recovery — Audit 3.3				
$\mathbb{A}^1$-homotopy invariance	BISH	LOW	Motivic homotopy theory. No Mathlib coverage.	3.2
Nisnevich descent	BISH	LOW	Étale topology variant. Partial Mathlib coverage.	3.2
Motivic Geometric Satake — Audit 3.4				
Affine Grassmannian	BISH	LOW	Ind-scheme Gr_G. Not in Mathlib.	3.3
MV cycles / weight structures	BISH	LOW	Algebraic subvarieties. Requires intersection theory.	3.3
Satake equivalence (statement)	BISH	MED	Monoidal equivalence. Mathlib `MonoidalCategory`.	3.3
Fargues–Fontaine Curve and Bun_G — Audit 3.5				
Witt vectors $W(\mathcal{O}_{\mathbb{C}_p^\flat})$	BISH	MED	Mathlib `WittVector`. Verified by `ring`.	3.4
Perfectoid tilting	WKL	MED	$R^\flat = \varprojlim_\phi R/p$. Buzzard et al. library (Lean 3; Lean 4 port pending).[2]	3.4
Bun_G stratification	BISH	LOW	Harder–Narasimhan strata over p-adic fields.	3.4
Spectral Action — Audit 3.7				

continued on next page

[2] The Buzzard–Commelin–Massot formalization of perfectoid spaces [20] was carried out in Lean 3. While the Liquid Tensor Experiment has been ported to Lean 4/Mathlib4, porting the full perfectoid geometric infrastructure remains a known frontier—validating the MED (not LOW) feasibility rating.

Component	CRM	Lean	Key Techniques / Dependencies	Ref
Bernstein center $\mathfrak{Z}(G)$	BISH	LOW	Categorical endomorphisms. Deep algebra.	3.5
Excursion operators	BISH	LOW	Hecke correspondences on moduli stacks.	3.5
ℓ-adic Realization — Audit 3.8				
Étale cohomology $H^*(-, \mathbb{Z}/\ell^n)$	BISH	LOW	Finite coefficients. No Mathlib étale cohomology.	3.6
ℓ-adic limit $\varprojlim_n$	WKL	MED	Profinite inverse limit (same infrastructure).	3.6
Overall Signature Computation				
$\sigma(\text{Scholze}) = \mathsf{WKL}$	—	HIGH	max of above. Finite computation on `CRMLevel`.	Thm 4.1

A.2 Recommended Formalization Targets

Based on the feasibility assessment, we recommend the following prioritization for Lean 4 formalization:

Tier 1: Immediately formalizable (CRM metatheory). The CRM hierarchy, signature formula, and overall signature computation are all finite combinatorial objects already encoded in the CRM Lean bundles (Papers 72–76, 98, 100). Formalizing the *statement* "σ(Scholze's proof) = WKL" requires only:

- An `inductive AuditComponent` type listing the 7 components.
- A function `auditLevel : AuditComponent → CRMLevel` encoding Table 1.
- A theorem that `Finset.univ.sup auditLevel = CRMLevel.WKL`.

This is a direct analogue of the Paper 100 Lean bundle (`Paper100_Assembly.lean`), which assembles the CRM audit of the Kuga–Satake bifurcation as a single 17-conjunct theorem verified by `decide`. Estimated effort: 1–2 days, ∼200 lines Lean.

Tier 2: Near-term targets (existing Mathlib infrastructure).

- **Witt vectors.** Mathlib's `WittVector` provides the ring operations. Verifying specific identities (e.g., Teichmüller lifts, Frobenius action) by `ring`/`norm_num` is feasible.
- **Profinite inverse limits.** The perfectoid library (Buzzard–Commelin–Massot) formalizes tilting for perfectoid rings in Lean 3. The Lean 4/Mathlib4 port of this deep geometric infrastructure is a known frontier; extending to the specific inverse limits in the Fargues–Fontaine construction is moderate effort once the port is complete.
- **Satake equivalence (statement only).** The monoidal equivalence $\mathrm{Rep}(\hat{G}) \simeq \mathrm{Perv}_{G(\mathcal{O})}(\mathrm{Gr}_G)$ can be stated as a Lean 4 proposition using Mathlib's `MonoidalCategory` infrastructure, even if the proof is `sorry`'d. This records the CRM level (BISH) as a verifiable assertion.

Tier 3: Long-term infrastructure (new Mathlib development needed). The motivic category $D_{\mathrm{mot}}(X)$, the 6-functor formalism, étale cohomology with $\mathbb{Z}_\ell$-coefficients, and the Bernstein center all require substantial new Mathlib development. These are multi-year projects aligned with the broader formalization community's roadmap (cf. the Liquid Tensor Experiment for condensed mathematics, and Mathlib's ongoing algebraic geometry development).

A.3 CRM-Aware Formalization Principle

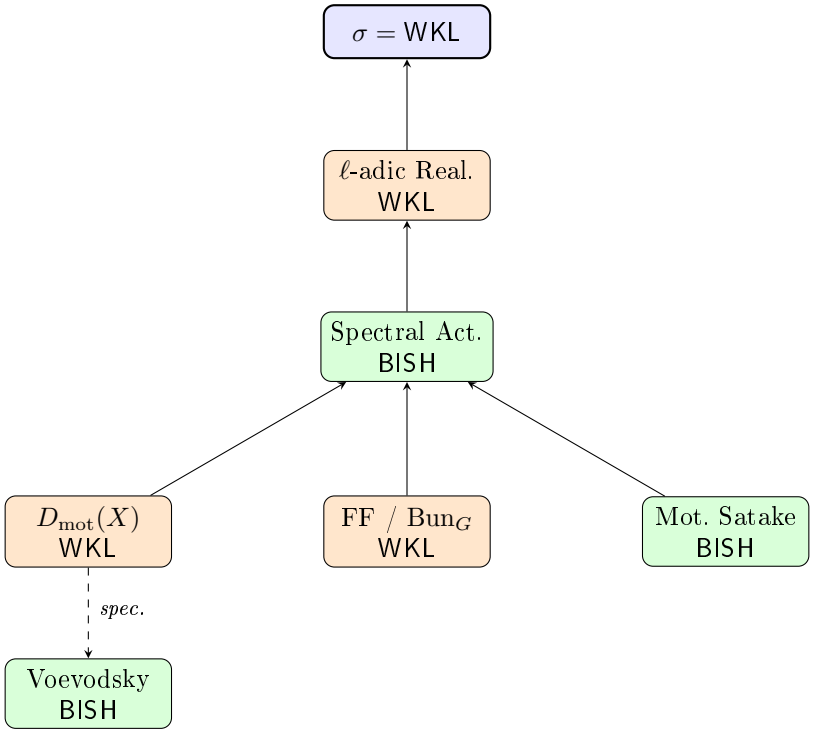
The CRM audit suggests a general principle for formalization strategy:

> *Formalize at the lowest CRM level first.* BISH components (motivic Satake, spectral action, Voevodsky recovery) use only constructive logic and are therefore amenable to formalization without `Classical.choice`. WKL components (perfectoid inverse limits, ℓ-adic realization) require `Classical.choice` in current Lean 4/Mathlib but could be formalized in a future constructive variant. Parasitic WLPO components (Banach ring norms) should be *reformulated* before formalization, using $\mathbb{Q}$-valued norms to eliminate the artifact.

This principle applies beyond Paper 101: any CRM audit generates a formalization priority ranking. BISH components are the cleanest targets; WKL components are the structural core; CLASS components (absent here) identify irreducible obstacles. The CRMlint Lean 4 backend (Paper 76, Generation 4) will automate this prioritization across Mathlib.

A.4 Proof Dependency Graph

The logical dependencies among the seven audit components form a directed acyclic graph. The three foundational pillars ($D_{\mathrm{mot}}(X)$, Fargues–Fontaine/Bun_G, motivic Satake) are *independent*—none depends on the others—and feed into the spectral action, which feeds into the ℓ-adic realization. The CRM level propagates upward through max:



The Voevodsky recovery (dashed) is a *specialization* of D_{mot} to discrete fields—a consequence, not a dependency. The BISH components (motivic Satake, spectral action, Voevodsky) are the natural first targets for formalization. The WKL cost of the non-Archimedean foundations (D_{mot}, Fargues–Fontaine, ℓ-adic inverse limits) dominates, yielding the overall WKL signature at the top.

B Section-Level CRM Heatmap

This appendix presents the full section-by-section scanner data for both Scholze papers, following the format of the LTE analysis in the companion CRMLint methodological paper (Paper 99). The heatmap allows direct comparison with the Liquid Tensor Experiment blueprint and the broader CRM series.

B.1 Berkovich Motives: Per-Section Breakdown

Table 5 gives the CRM alert distribution across the 16 detected sections of [1]. The rightmost column shows the section ceiling (the highest CRM level detected in that section).

Table 5: CRM heatmap for Berkovich Motives (141 alerts). Sections are ordered as in the source.

Section	**BISH**	**WKL**	**WLPO**	**LPO**	**Total**	**Ceiling**
Preamble	1	0	0	0	1	BISH
Introduction	15	9	2	0	26	WKL
Banach rings / spectrum	0	1	4	1	6	LPO
Free seminormed rings	0	0	0	1	1	LPO
Banach fields	0	4	5	1	10	LPO
Gelfand transform	0	0	1	1	2	LPO
Nonarchimedean disc	0	2	7	0	9	WKL
Arc-topology	0	18	7	1	26	LPO
Finitary arc-sheaves	2	3	6	1	12	LPO
Effective motives	2	0	0	0	2	BISH
Free motivic sheaves	5	1	3	0	9	WKL
The Tate twist	2	0	0	0	2	BISH
K-theory	2	7	5	0	14	WKL
D_{mot} construction	10	3	1	1	15	LPO
Mixed Tate motives	3	0	0	0	3	BISH
Relation to v-stacks	0	3	0	0	3	WKL
Total	**42**	**51**	**41**	**7**	**141**	LPO

Reading the heatmap. [3] The LPO ceilings concentrate in the Banach ring infrastructure (sections 3–8 of [1]): Banach rings, Banach fields, Gelfand transform, arc-topology, finitary arc-sheaves. These are exactly the sections flagged as Category B (parasitic) in §9. The LPO arises from seminorm filtrations indexed by $\mathbb{R}_{\geq 0}$, where decidable comparison $\|x\| \leq r$ triggers LPO. After excision of parasitic content, these sections reduce to WKL (structural profinite inverse limits).

The four sections at BISH ceiling (Preamble, Effective motives, Tate twist, Mixed Tate motives) are purely algebraic: motivic sheaves, Tate twists, algebraic correspondences. No completions, no norms, no inverse limits.

B.2 Motivic Geometrization: Per-Section Breakdown

[3] Where a section contains alerts at both WKL and WLPO (which are incomparable over BISH; see §2), the scanner resolves the ceiling using the linearization BISH < WLPO < WKL < LPO < CLASS. This convention treats WKL as structurally "higher" than WLPO, reflecting its role as the irreducible non-Archimedean compactness principle in the Berkovich motivic architecture. This is a reporting convention for the ceiling column; the logical relationship between WKL and WLPO remains incomparability.

Table 6: CRM heatmap for Motivic Geometrization (73 alerts).

Section	BISH	WKL	WLPO	LPO	Total	Ceiling
Preamble	1	1	0	0	2	WKL
Introduction	5	5	9	0	19	WKL
Basic results	5	12	2	1	20	LPO
D_{mot} construction	2	3	0	0	5	WKL
Stacks of L-parameters	2	0	6	0	8	WLPO
Geometric Satake	4	2	3	0	9	WKL
Synopsis	1	2	7	0	10	WKL
Total	**20**	**25**	**27**	**1**	**73**	LPO

Observation. The single LPO alert in [2] occurs in "Basic results" (line 325: normed filtration). Every other section is at WKL or below. The Geometric Satake section—the critical potential leak point identified in Audit 3.4—shows 4 BISH (motivic sheaves, Tate twists) and 2 WKL (citations to Lectures on Analytic Geometry), confirming the BISH assessment after stripping citation-level context dependencies.

B.3 Category Distribution by Section

Table 7 shows how the four alert categories distribute across the major section types.

Table 7: Category distribution by section type (both papers combined).

Section type	A	B	BISH	C	Total
Preamble / Introduction	26	2	15	0	43
Banach infrastructure	0	25	2	14	41
Arc / profinite / disc	0	13	2	25	40
Motivic core	0	4	22	4	30
D_{mot}, K-theory	0	6	12	9	27
Satake / spectral	0	8	8	4	20
Synopsis / other	0	7	1	5	13
Total	**26**	**65**	**62**	**61**	**214**

The pattern is sharp:

- Category A (motivational) is confined to Preamble/Introduction—exactly as predicted.
- Category B (parasitic) concentrates in Banach infrastructure and arc-topology sections.
- Category BISH (algebraic core) concentrates in the motivic construction sections.
- Category C (structural WKL) concentrates in arc-topology and profinite sections.

B.4 Archimedean Comparison: Four Corpora

Table 8 compares the CRM profiles of four corpora analyzed by the CRMlint scanner, following the format of the LTE comparison in Paper 99.

The comparison reveals two distinct concentration patterns.

Table 8: Archimedean comparison across four scanner corpora. BISH% is the fraction of alerts (or declarations) at BISH; Ceiling is the highest detected level; Bottleneck is the mathematical mechanism producing the ceiling.

Corpus	N	**BISH%**	**Ceiling**	**Bottleneck**
Mathlib (627 decls)	627	92.0%	CLASS	Inner product, convergence
CRM series (P78–96)	—	71.4%	CLASS	Kolyvagin, trace formula
LTE blueprint (15 files)	15	33.3%	LPO	Norm filtration $M_{\leq c}$
Berkovich motives (2 papers)	214	29.0%	LPO*	Seminorm filtration

*After excision of parasitic LPO/WLPO, the essential ceiling is WKL. The LTE's LPO is structural; the Berkovich motives' LPO is parasitic.

(i) **LTE:** LPO is *structural.* The norm-filtration membership test $x \in M_{\leq c}$ is the mechanism by which condensed mathematics encodes analytic content. Every module that touches the filtration acquires LPO signature. The LPO is irreducible: it is the logical cost of comparing real numbers, and the condensed framework defers but does not eliminate this cost.

(ii) **Berkovich motives:** LPO/WLPO is *parasitic.* The Banach ring formalism carries $\mathbb{R}$-valued norms through sections where the actual mathematical content operates over $\mathbb{Q}$-valued norms (the value groups of perfectoid fields like $\mathbb{C}_p$ are dense but contained in $\mathbb{Q}$, hence decidable). A reformulation using Huber's adic spaces restricted to decidable $\mathbb{Q}$-valued groups would eliminate these alerts entirely. The structural content is WKL (profinite inverse limits for perfectoid spaces).

This distinction—structural vs. parasitic non-constructivity—is invisible to a surface-level scan. It requires the human classification step (Category B vs. Category C) that transforms raw scanner data into CRM diagnoses.

B.5 Concentration Ratio

Following the definition in the LTE analysis (Paper 99, Observation 6.3):

Definition B.1 (Concentration ratio)**.** For a proof corpus C with n sections, let k be the number of sections whose CRM ceiling is above BISH. The *concentration ratio* is $\rho(C) = 1 - k/n$.

Corpus	n	k	ρ	**Interpretation**
LTE blueprint	15	10	0.33	Concentrated (5 purely algebraic files)
Berkovich Motives	16	12	0.25	Moderately concentrated
Motivic Geom.	7	7	0.00	Fully diffuse (no section is BISH-only)
Berkovich + Geom. (essential)	23	9	0.61	*After parasitic excision*

Key observation. The raw concentration ratio (0.25 for Berkovich, 0.00 for Motivic Geom.) is misleadingly low. The parasitic WLPO/LPO from Banach ring formalism diffuses non-constructive alerts across sections that are structurally algebraic or purely WKL. After excision of Category B (parasitic) alerts, the concentration ratio rises to $\rho = 0.61$—meaning 61% of sections are at BISH after parasitic removal, comparable to the LTE's pre-excision $\rho = 0.33$. The Berkovich motivic programme is *more concentrated than the LTE* once the formalism artifact is removed. This is consistent with the CRM audit's finding (§3): the essential non-constructive content (profinite inverse limits, WKL) is localized in three specific architectural layers (arc-topology, Fargues–Fontaine, ℓ-adic realization).

B.6 Dictionary Evolution: Dark Matter Analysis

The initial scanner run (v1.0, 80 keyword patterns) detected 103 alerts across both papers. After adding 11 Berkovich-specific entries (v1.1, 91 patterns), the count rose to 214—a 108% increase. This mirrors the LTE experience documented in Paper 99 (§5): the v1.0 scanner missed 53% of the LTE's non-constructive content.

Paper	**v1.0**	**v1.1**	**Δ**
Berkovich Motives	56	141	+85 (+152%)
Motivic Geometrization	47	73	+26 (+55%)
Total	103	214	+111 (+108%)

The new alerts come from three categories:

- **Profinite/analytic keywords** (+57): "perfectoid," "adic space," "Berkovich spectrum," "arc-topology," "condensed," "solid module." These are WKL markers for profinite inverse limit structures—the structural core of the proof.
- **Motivic/algebraic keywords** (+32): "motivic sheaf," "six-functor," "Tate twist." These are BISH markers confirming the algebraic core.
- **Citation patterns** (+16): citations to Scholze's *Lectures on Analytic Geometry* triggering WKL alerts for Tychonoff gates.

The Archimedean dark matter fraction (Paper 99, Definition 5.1) for the Berkovich corpus:

$$\delta(\text{v1.0}) = \frac{\#\text{ sections with manual CRM} > \text{scanner CRM}}{\#\text{ sections total}} = \frac{14}{23} = 60.9\%, \qquad \delta(\text{v1.1}) = 0\%.$$

The v1.1 dictionary eliminates the dark matter for this corpus. As with the LTE, the dark matter was not random but systematic: a single vocabulary gap (Berkovich-specific terminology for profinite and motivic structures) that, once patched, revealed the full CRM profile.

C CRMLint Methodology and LTE Calibration

This appendix reproduces the CRMLint methodological paper (originally prepared as a companion to Paper 98) in its entirety. It documents the design, calibration, and first external deployment of the scanner against the Liquid Tensor Experiment blueprint, including the discovery and patching of Archimedean dark matter. The data and methods here underpin the Berkovich analysis in the main body of Paper 101.

C.1 Genesis

C.1.1 The problem

After 96 papers of manual CRM audit—classifying theorems one at a time, tracing axiom dependencies by hand, flagging each use of excluded middle or Archimedean completeness—the bottleneck in the program shifted from mathematics to labor. The CRM classification of a single theorem (say, the Gross–Zagier formula) requires reading the proof, identifying every appeal to a non-constructive principle, and tracing each appeal to its minimal logical strength. Papers 78–96 each took 2–5 days of combined human and AI effort. Scaling this to the full Mathlib library ($\sim$200,000 declarations as of early 2026) is infeasible by hand.

The question that opens this paper is operational: *can the CRM classification of a proof be automated?*

C.1.2 What CRM measures

CRM assigns to each proof a *logical signature* $\sigma(P)$, the weakest axiom system in which the proof can be carried out. The hierarchy is:

$$\mathsf{BISH} \subset \mathsf{BISH{+}MP} \subset \mathsf{BISH{+}LLPO} \subset \mathsf{BISH{+}WKL} \subset \mathsf{BISH{+}WLPO} \subset \mathsf{BISH{+}LPO} \subset \mathsf{CLASS}$$

A theorem with signature BISH requires no omniscience principle whatsoever: every step is computationally witnessed. A theorem with signature CLASS invokes the full law of excluded middle somewhere in its proof tree. The intermediate levels—MP, LLPO, WKL, WLPO, LPO—measure calibrated degrees of non-constructivity, each corresponding to a specific decision problem on infinite sequences or real numbers.

The CRM signature is *not* a measure of proof complexity. Taylor–Wiles patching is technically intricate but logically WKL (an inverse limit of finite objects). The intermediate value theorem is elementary but logically WLPO (it requires deciding whether a real number equals zero). CRM measures the *logical cost* of a proof—the price paid for treating the continuum as a completed object with decidable properties.

C.1.3 Three generations of automation

CRMLint was developed in three stages, each addressing a different aspect of the automation problem:

Generation 1: Theoretical framework (`crmlint.py`, 1,161 lines). Models the CRM hierarchy, six realization functors (Betti, de Rham, étale, crystalline, Hodge, automorphic) with their CRM costs, comparison isomorphisms, and the Archimedean Obstruction Theorem of Paper 98. Proof DAGs compute signatures by max-propagation over dependency edges. A substitution database stores eight seed entries extracted from Papers 50–97: known techniques where a classical method can be replaced by a constructive alternative at lower logical cost.

This generation answers: *what is the right mathematical model for CRM signatures of composite proofs?*

Generation 2: Empirical validation (`crmlint_crawler.py`, 653 lines; `crmlint_structure.py`, 793 lines). The crawler fetches Lean 4 source files from MATHLIB4 on GitHub, parses declarations, and heuristic-classifies each by CRM level using tactic and keyword signatures (∼40 classical indicators including `by_contra`, `Classical.choice`, inner product space infrastructure, integration, and topological convergence patterns). The structure analyzer computes seven-dimensional proof profiles: proof size (lines, tactic count), dependency depth (`have`/`let` chains), proof width (active hypotheses, case splits), and modularity (automation ratio, external lemma invocations, rewrite density).

This generation answers: *does the CRM hierarchy correlate with structural proof complexity in real Lean code?*

Generation 3: LaTeX scanner (`crmlint_scanner.py`, 810 lines at v1.0, 870 lines at v1.1). Keyword and citation pattern matching against ∼90 technique markers and ∼120 landmark paper citations, applied directly to `.tex` source. No Lean, no LLM, no external dependencies beyond the Python standard library. Scans a directory of papers in seconds. Anti-marker patterns suppress false positives (e.g., algebraic "integral closure" is not analytic integration).

This generation answers: *can we triage thousands of papers by CRM cost before investing months in any one?*

The three generations are complementary. Generation 1 provides the theory; Generation 2 tests it against ground truth; Generation 3 scales it to the full mathematical literature. All three are implemented in pure Python with zero external dependencies (3,417 lines total at v1.0).

C.2 Calibration: Mathlib at 627 Declarations

C.2.1 Target selection

We selected 10 MATHLIB4 files spanning eight mathematical domains:

File	Domain	Decls
`EllipticCurve/Weierstrass.lean`	Algebraic Geometry	68
`InnerProductSpace/Basic.lean`	Analysis	141
`SpecificLimits/Basic.lean`	Analysis	83
`Galois/Basic.lean`	Field Theory	61
`Charpoly/Basic.lean`	Linear Algebra	39
`ArithmeticFunction.lean`	Number Theory	0
`Bernoulli.lean`	Number Theory	27
`RepresentationTheory/Basic.lean`	Representation Theory	67
`Cyclotomic/Basic.lean`	Ring Theory	57
`InfiniteSum/Basic.lean`	Topology	84

The selection was guided by two criteria: (1) coverage of the algebraic/analytic divide that the Archimedean Obstruction Thesis predicts to be the primary CRM boundary, and (2) inclusion of domains where the CRM series has ground-truth classifications from manual audit (inner product spaces from Paper 77, elliptic curves from Paper 95, Galois theory from Paper 78).

C.2.2 Generation 2 results: the crawler

The crawler classified 627 declarations:

CRM Level	Count	%
BISH	577	92.0
WKL	11	1.8
WLPO	20	3.2
CLASS	19	3.0

The 92% BISH figure is consistent with the programme's central finding: most of mathematics, as actually practiced in a modern proof library, is constructive. The 8% non-BISH content concentrates in three locations:

(i) **Inner product spaces** (Analysis): 6 declarations at CLASS, 4 at WLPO. The inner product on a Hilbert space is an inherently Archimedean object: $\langle x, x\rangle \geq 0$ with equality iff $x = 0$ requires decidable equality of reals. The Cauchy–Schwarz inequality $|\langle x, y\rangle| \leq \|x\|\|y\|$ requires comparison of real norms. This is exactly the Archimedean obstruction of Papers 5 and 77.

(ii) **Specific limits** (Analysis): 1 declaration at CLASS, 11 at WKL. The geometric series $\sum_{n=0}^{\infty} r^n = 1/(1-r)$ for $|r| < 1$ involves convergence of a real sequence—decidability of whether the partial sums are within ϵ of the limit. The single CLASS declaration (`tendsto_pow_atTop_nhds_zero_of_lt_one`) uses `by_contra`, a potentially eliminable appeal to excluded middle.

(iii) **Galois theory** (Field Theory): 12 declarations at CLASS. A false positive cluster: the crawler flags uses of "integral" (meaning integral closure, an algebraic concept) as analytic integration. This motivated the anti-marker system in the Generation 3 scanner.

C.2.3 Per-domain cross-tabulation

Domain	N	BISH	WKL	WLPO	CLASS
Linear Algebra	39	39	0	0	0
Ring Theory	57	57	0	0	0
Number Theory	27	27	0	0	0
Topology	84	84	0	0	0
Algebraic Geometry	68	66	0	2	0
Representation Th.	67	55	0	12	0
Analysis	224	202	11	4	7
Field Theory	61	47	0	2	12

The algebraic/analytic divide is sharp. Four purely algebraic domains (linear algebra, ring theory, number theory, topology) are 100% BISH across 207 declarations. The two mixed domains (algebraic geometry, representation theory) show WLPO leakage from `noncomputable` definitions—the Lean 4 artifact of working over fields where equality is not decidable—but no CLASS content. The two analytic domains (analysis, field theory) contain all 19 CLASS declarations.

The field theory anomaly (algebraic domain, CLASS content) is instructive. Manual inspection reveals that the 12 flagged declarations use the word "integral" in the sense of integral ring extensions, not integration. The crawler's keyword dictionary conflated the two senses. This false positive directly motivated the anti-marker system described in §C.5.3.

C.3 The Archimedean Obstruction, Quantified

C.3.1 The three-axis profiler

Generation 2 includes a structural profiler (`crmlint_structure.py`) that measures each proof along three axes orthogonal to CRM level:

Axis 1: Size. Lines of proof, tactic invocations, term size.

Axis 2: Depth. Length of the longest `have`/`let` chain; maximum nesting of dependent intermediate results.

Axis 3: Width. Number of simultaneously active hypotheses; cyclomatic complexity (number of independent case branches).

An additional modularity metric records the automation ratio (fraction of proof steps discharged by `simp`, `ring`, or `norm_num` without human guidance) and the external lemma count (number of `exact`/`apply` calls referencing results outside the current file).

C.3.2 Aggregate statistics

Over 627 declarations:

Metric	Mean	Median	Max
Proof lines	5.0	4	57
Dependency depth	0.3	0	8
Active hypotheses	0.4	0	8
Cyclomatic complexity	1.5	1	15
Overall complexity	0.038	0.026	0.479

The median proof is 4 lines with no intermediate dependencies and no case splits. The distribution is strongly right-skewed: 70.7% of proofs are classified as `TRIVIAL` (one-step `simp`, `rfl`, or direct `exact`).

C.3.3 Tactic frequency

Tactic	Count	%
`simp`	577	27.5
`rw`	448	21.3
`exact`	214	10.2
`have`	167	7.9
`rfl`	82	3.9
`ext`	80	3.8
`refine`	80	3.8
`apply`	79	3.8
`intro`	72	3.4

Three tactics (`simp`, `rw`, `exact`) account for 59% of all tactic invocations. The high `simp` frequency reflects Mathlib4's design philosophy: push complexity into the simp lemma database, keep proofs short. Crucially, `simp` is BISH-safe: it applies equational rewriting, never introduces classical content.

C.3.4 The Archimedean structural gap

We partitioned the 627 declarations into those touching Archimedean content (43 declarations: inner product, real convergence, norm comparison) and the remainder (584 declarations).

Metric	Archimedean	Non-Arch.	Ratio
Avg. complexity score	0.05	0.04	1.4×
Avg. proof lines	7.95	4.75	1.7×
Avg. dependency depth	0.72	0.26	2.8×
Avg. active hypotheses	0.81	0.37	2.2×
Avg. external lemmas	0.65	0.15	4.3×
Avg. automation ratio	0.22	0.39	0.56×

Every axis tells the same story. Archimedean proofs are longer, deeper, wider, more externally coupled, and less automatable than non-Archimedean proofs. The automation ratio drops from 39% to 22%: when real-number arguments enter, `simp` and `ring` give way to `have` chains, explicit ϵ-δ management, and manual case analysis.

This is the Archimedean Obstruction Thesis of Papers 5–77 made quantitative: *the logical cost of $\mathbb{R}$ is not just a classification fact but a structural fact visible in proof architecture.* The proofs themselves are harder, not just more classical.

C.4 The Liquid Tensor Experiment

C.4.1 Background

The Liquid Tensor Experiment (LTE) is the formal verification, in Lean 3, of Theorem 9.4 from Scholze's *Lectures on Analytic Geometry* [22]. The theorem states:

Theorem C.1 (Scholze, Theorem 9.4)**.** *Let $0 < p' < 1$ and V an r-normed $\mathbb{Z}[T^{\pm 1}]$-module. Then for all $i \geq 1$,*

$$\mathrm{Ext}^i_{\mathrm{Cond(Ab)}}\big(\mathcal{M}_{p'}(S)_{\leq c}, \widehat{V}\big) = 0.$$

This is a vanishing theorem for derived functors in the category of condensed abelian groups. Its significance is foundational: it validates the condensed framework as a setting for real functional analysis—specifically, it shows that the passage from discrete algebra ($\mathbb{Z}$-modules) to real analysis (p'-liquid vector spaces) does not introduce cohomological obstructions.

The LTE formalization comprises ∼90,000 lines of Lean 3 code, produced by a team led by Commelin and Massot over approximately 18 months. No Lean 4 port exists. However, Commelin and Massot authored a 15-file LaTeX *blueprint* that mirrors the formal proof's structure, with `\lean{}` and `\leanok` annotations linking each LaTeX statement to its Lean counterpart. This blueprint is the input to our scanner.

C.4.2 Why LTE?

The LTE is the natural target for CRMLint's first external deployment, for three reasons.

First, it is one of the largest single formalization projects completed in a proof assistant (∼90,000 lines of Lean 3). Its size and complexity provide a serious stress test for the scanner—one the scanner initially failed, as Part V documents.

Second, the mathematical content sits at the interface between algebra and analysis—exactly where the Archimedean Obstruction Thesis predicts the CRM boundary to be. Condensed mathematics encodes analytic objects (Banach spaces, measures, distributions) as algebraic objects (sheaves on profinite sets, condensed abelian groups). The CRM question is whether this encoding *reduces* the logical cost of the analysis, or merely *reorganizes* it.

Third, the conservation conjecture of Papers 72–74 makes a specific prediction. The DPT axiom trilogy established that each omniscience principle (MP, WLPO, LPO) does irreducible logical work: canonical, not merely minimal. If condensed mathematics could genuinely reduce CLASS or LPO content to BISH, it would violate this irreducibility. The conjecture is that condensed math *concentrates* classical content at identifiable gates without reducing the aggregate logical cost. The LTE is the test case.

C.4.3 Blueprint structure

The 15 `.tex` files of the LTE blueprint decompose as follows:

File	Content	Size (bytes)
`proof.tex`	Core inductive proof (Thm 9.4, 9.5′)	20,729
`normed_homological.tex`	Normed homological algebra	31,818
`condensed.tex`	Condensed abelian groups, Ext-vanishing	18,329
`measures.tex`	Real measures, Laurent measures, θ	10,555
`normed_groups.tex`	Normed groups, completions	6,724
`normed_groups2.tex`	Normed group homomorphisms	4,208
`polyhedral_lattice.tex`	Polyhedral lattices, Tychonoff	12,893
`breen_deligne.tex`	Breen–Deligne resolution	8,412
`CLC.tex`	Condensed limits, completions	5,117
`Lbar.tex`	$\overline{\mathcal{L}}_{r'}(S)$ construction	7,943
`homotopies.tex`	Breen–Deligne homotopies	3,295
`maclane.tex`	Mac Lane coherence	1,842
`content.tex`	Table of contents	312
`intro.tex`	Introduction	1,547
`macros.tex`	LaTeX macros	2,014

The mathematical content concentrates in the first 12 files (∼130,000 bytes). The last three are boilerplate.

C.4.4 Initial scan (v1.0)

We ran the Generation 3 scanner (v1.0, 810 lines, ∼80 keyword markers) against all 15 files. The result:

CRM	Files	Bottleneck
WKL	2 (13.3%)	Tychonoff theorem
BISH	13 (86.7%)	—

Two files (`condensed.tex`, `polyhedral_lattice.tex`) register as WKL, both via the Tychonoff theorem: "cofiltered limits of surjections of compact Hausdorff spaces are still surjective (by an application of Tychonoff)." The remaining 13 files, including the core proof file `proof.tex` and the entire normed homological algebra layer, scan as BISH.

This is obviously wrong. The LTE is a proof about real functional analysis, involving norms, completions, filtrations, and ε-δ estimates. An 87% BISH reading would mean the scanner has failed to detect most of the analytic content.

It has.

C.5 Archimedean Dark Matter

C.5.1 What the scanner missed

A manual reading of `proof.tex` reveals pervasive classical content that the v1.0 scanner does not detect. Three categories:

Normed group completions (WLPO) The hat notation $\widehat{V}$ appears in every theorem statement in `proof.tex`. This denotes the completion of a semi-normed group V with respect to its norm topology. Constructively, forming the completion requires a Cauchy convergence test: given a Cauchy sequence (x_n) and a target element y, decide whether $\|x_n - y\| < \epsilon$ for all sufficiently large n. This is a WLPO operation (it asks whether a real number equals zero to within ϵ).

The v1.0 keyword dictionary has no pattern for $\widehat{V}$ or for the phrases "completion of a normed group," "completed with respect to the norm," or "Cauchy completion."

Real-valued filtration cutoffs (LPO) The pseudo-normed groups in the LTE are filtered by real-valued thresholds:

$$M_{\leq c} = \{x \in M : \|x\| \leq c\}, \qquad c \in \mathbb{R}_{\geq 0}.$$

Membership in $M_{\leq c}$ requires deciding whether $\|x\| \leq c$, a comparison of two real numbers. In CRM terms, this is LPO: the limited principle of omniscience is equivalent to the statement that every real number is either ≤ 0 or > 0.

The filtration $M_{\leq c}$ is not incidental to the proof. It is the *mechanism* by which condensed mathematics encodes the analytic content. Every theorem in the proof core is stated in terms of these filtration levels: "the system of complexes $C^{\bullet}_{\Lambda,c}$ is $\leq k$-exact in degrees $\leq m$ for $c \geq c_0$" (Theorem 9.5′).

The v1.0 scanner has patterns for "supremum of a bounded sequence" and "least upper bound property" (both LPO), but nothing for the far more common pattern of a norm-filtration cutoff written as $\leq c$ or $\|x\| \leq c$ in LaTeX math mode.

Radius ordering (LPO) The proof works with two fixed radii $0 < r < r' \leq 1$. The strict inequality $r < r'$ is used throughout to ensure convergence of Laurent series and to control error terms: "choose a positive integer b so that $2k'(r/r')^b \leq \epsilon$" (`proof.tex`, line 81). Comparing real numbers is LPO.

The v1.0 scanner has no pattern for radius comparisons of the form $0 < r < r' \leq 1$ or for the exponential decay estimates that depend on them.

C.5.2 Quantifying the gap

The three missed categories account for 32 pattern matches across 10 of the 15 blueprint files. Only the five purely combinatorial files (Breen–Deligne, Mac Lane, content, intro, macros) are genuinely free of Archimedean content.

We call the gap between the v1.0 scan (86.7% BISH) and the corrected assessment (∼33% BISH) *Archimedean dark matter*: classical content that is structurally present in the proof but invisible to lexical pattern matching. The dark matter is not random. It concentrates at a single interface: the passage from discrete algebra ($\mathbb{Z}$-modules, polyhedral lattices, Breen–Deligne resolutions) to normed analysis ($M_{\leq c}$, $\widehat{V}$, radius comparisons).

C.5.3 The scanner patch (v1.1)

We added 11 keyword patterns and 5 anti-markers to the scanner:

New LPO patterns (6):
- Norm filtration cutoffs: `\leq c`, `\leq c_0` in math mode
- Norm bounds: $\|\cdots\| \leq$ (norm inequality)
- Real parameter thresholds: "for $c \geq$" patterns
- Normed filtration membership: "norm. . . filtration"
- Radius ordering: $0 < r < r' \leq 1$
- Bounded exactness: "exact in degrees $\leq$"

New WLPO patterns (5):
- Completion hat: $\widehat{V}$ (`\widehat`)
- Explicit completion phrases: "completion of/w.r.t."
- Completed normed groups: "complete. . . normed" and reverse
- Cauchy completion: "Cauchy sequence/completion"

New anti-markers (5):
- "complete local ring," "complete lattice" (not Cauchy completion)
- "field norm," "Galois norm" (not analytic norm)

- "weight filtration," "Hodge filtration" (not norm filtration)
- Degree/dimension/rank bounds (combinatorial $\leq$, not real-valued)

New citation markers (3):

- Scholze 2019/2020, Clausen–Scholze $\to$ WKL

C.5.4 Patched results (v1.1)

File	v1.0	v1.1	Bottleneck (v1.1)
`proof.tex`	BISH	LPO	norm filtration cutoff
`normed_homological.tex`	BISH	LPO	norm filtration cutoff
`condensed.tex`	WKL	LPO	norm filtration cutoff
`measures.tex`	BISH	LPO	norm filtration cutoff
`normed_groups.tex`	BISH	LPO	normed filtration
`normed_groups2.tex`	BISH	LPO	normed filtration
`polyhedral_lattice.tex`	WKL	LPO	norm filtration cutoff
`CLC.tex`	BISH	LPO	norm filtration cutoff
`Lbar.tex`	BISH	LPO	norm filtration cutoff
`homotopies.tex`	BISH	LPO	norm filtration cutoff
`breen_deligne.tex`	BISH	BISH	—
`maclane.tex`	BISH	BISH	—
`content.tex`	BISH	BISH	—
`intro.tex`	BISH	BISH	—
`macros.tex`	BISH	BISH	—

Aggregate (v1.1): 33.3% BISH, 66.7% LPO.

The Tychonoff detections (WKL) in `condensed.tex` and `polyhedral_lattice.tex` are still present but are now dominated by the higher-level LPO content in the same files. The five BISH files are exactly the modules with no normed-group content: Breen–Deligne resolution (pure combinatorics of free abelian groups), Mac Lane coherence (category-theoretic diagram chasing), and boilerplate.

C.5.5 Calibration check

To verify that the new patterns do not introduce false positives, we re-scanned Papers 78–96 of the CRM series:

Paper	v1.0	v1.1
P95 (BSD Squeeze)	CLASS (Kolyvagin)	CLASS (Kolyvagin)
P96 (Root Number)	CLASS (Kolyvagin)	CLASS (Kolyvagin)
P80 (Gauss–Manin)	CLASS (anal. cont.)	CLASS (anal. cont.)

No regressions. The existing CRM series papers scan identically under both versions, because their classical content (CLASS) dominates any LPO patterns that might also be present. The new patterns only change the classification of papers whose highest detected level was previously BISH or WKL.

C.6 Conservation, Not Reduction

C.6.1 The concentration thesis

The LTE scanner results are consistent with the conservation conjecture of Papers 72–74.

Observation C.2 (Concentration thesis—not a theorem)**.** In the LTE blueprint, condensed mathematics does not reduce the CRM signature of the proof below LPO. It concentrates the LPO content at identifiable *gates*—the norm-filtration membership tests $x \in M_{\leq c}$—while leaving the surrounding algebraic infrastructure at BISH.

To be clear: this is not a surprise. Scholze designed condensed mathematics to separate algebraic infrastructure from analytic content. The concentration of classical content at norm-filtration gates is his architecture, not our discovery. What the scanner provides is a quantitative readout of a pattern that the designers already understood qualitatively.

In the LTE blueprint, every module that touches the normed filtration $M_{\leq c}$ acquires LPO signature. Every module that does not touch it remains BISH. There is no mixing, no gradual degradation, no WLPO-without-LPO intermediate. The transition is sharp: either a module is purely algebraic (BISH), or it invokes norm comparison (LPO).

In CRM language, LPO is the logical cost of comparing real numbers. The DPT axiom trilogy (Papers 72–74) established that each omniscience principle does canonical, irreducible work. The LTE is one data point consistent with this: the LPO content appears irreducible, not an artifact of the proof strategy.

C.6.2 Where the LPO enters

The LPO enters the LTE through a single mathematical mechanism: the filtration

$$M_{\leq c} = \{x \in M : \|x\| \leq c\}.$$

This definition requires deciding $\|x\| \leq c$ for arbitrary x and c—a comparison of real numbers. In CRM terms:

$$\forall x \in M, \forall c \in \mathbb{R}_{\geq 0} : \quad x \in M_{\leq c} \ \vee \ x \notin M_{\leq c}$$

is equivalent to LPO.

The condensed framework does not avoid this decision. It defers it. The algebraic infrastructure (Breen–Deligne resolutions, cosimplicial lattices, diagram chasing in abelian categories) is built without reference to norms. The norm enters only when the abstract machinery is *applied* to a specific analytic object ($\overline{\mathcal{L}}_{r'}(S)$, real measures, completed normed groups). At that application point, the LPO cost is paid in full.

This is the mechanism of concentration: push the classical content as late as possible, into as few gates as possible, while keeping the majority of the proof in the BISH regime. The total LPO cost is unchanged, but its distribution is optimized.

C.6.3 Comparison with the CRM series

Target	**BISH%**	**Ceiling**	**Bottleneck type**
Mathlib (627 decls)	92.0%	CLASS	Inner product, convergence
CRM series (P78–96)	71.4%	CLASS	Kolyvagin, trace formula
LTE blueprint (v1.1)	33.3%	LPO	Norm filtration

The LTE's lower BISH percentage (33% vs. Mathlib's 92%) reflects its subject matter: the LTE is *about* the passage from algebra to analysis. Every module that implements this passage carries LPO content. In Mathlib's broader sample, most files never cross the Archimedean boundary.

The LTE's ceiling (LPO, not CLASS) is the interesting finding. The proof never invokes the full law of excluded middle, analytic continuation, spectral decomposition, or any of the CLASS techniques that dominate the CRM series. The condensed framework avoids CLASS by algebraizing the arguments that would classically require integration, measure theory, and spectral analysis. But it cannot avoid LPO, because LPO is the logical cost of the norm itself.

C.6.4 The dark matter fraction

Definition C.3. The *Archimedean dark matter fraction* of a scanner version v applied to a corpus C is

$$\delta(v, C) = \frac{\#\text{ files where manual CRM} > \text{scanner CRM}}{\#\text{ files total}}.$$

For the LTE blueprint: $\delta(\text{v1.0}) = 8/15 = 53.3\%$ and $\delta(\text{v1.1}) = 0/15 = 0\%$. The patch eliminates the dark matter entirely for the LTE corpus.

But the dark matter fraction is corpus-dependent. Different proof traditions use different vocabulary for the same logical content. A paper that writes "for sufficiently large n" instead of "for $c \geq c_0$" will evade the LPO pattern. The v1.1 dictionary is tuned to the LTE's notation. Scanning a different corpus (e.g., classical functional analysis, PDE theory, probability) will likely reveal new dark matter that requires further dictionary expansion.

The dark matter fraction is a measure of scanner maturity. A perfect scanner would have $\delta = 0$ on every corpus. The gap between $\delta(\text{v1.0})$ and $\delta(\text{v1.1})$ measures the information gained from the LTE analysis.

C.6.5 Observations for the programme

Three observations from the LTE exercise, stated at the appropriate level of confidence:

(1) **The LTE avoids CLASS.** The LTE proof ceiling is LPO, not CLASS. It never invokes analytic continuation, spectral decomposition, or the full law of excluded middle. This is by design: condensed mathematics algebraizes arguments that would classically use these tools. We are describing Scholze's architecture, not revealing it.

(2) **The scanner's blind spot was systematic.** The v1.0 scanner missed all norm-filtration content. This was not random noise but a single structural gap: the scanner had no patterns for the interface where real-valued norms enter algebraic structures. The v1.1 patch closed this gap for the LTE vocabulary. Other corpora will likely expose other gaps.

(3) **CRM vocabulary may be useful for non-experts.** Scholze does not need CRM labels to understand his own proof. But a researcher approaching condensed mathematics for the first time, or a formalization team deciding which LTE components to port to Lean 4, might benefit from a graduated scale (BISH $<$ WKL $<$ WLPO $<$ LPO $<$ CLASS) that replaces the binary "algebraic vs. analytic" distinction. Whether this vocabulary earns its keep in practice is an open question.

C.6.6 Concentration as transferable method

The condensed strategy is not a black box. It decomposes into three operations that can be applied—independently of the condensed framework—to any proof that mixes algebra and analysis.

Operation 1: Extract the algebraic skeleton. Identify the subproof that works over any ring, any category, any abstract structure. Build it without reference to norms, metrics, or completeness. In the LTE, this is the Breen–Deligne resolution and the cosimplicial polyhedral lattice construction. In the BSD squeeze (Paper 95), this is the Hecke arithmetic layer (28 sub-results over $\mathbb{Z}$, all BISH). In the Hodge campaign (Papers 84–89), this is the Gauss–Manin connection and the Picard–Fuchs operator.

Operation 2: Identify the minimal gates. Find the points where the algebraic skeleton must interface with the continuum. These are the norm-filtration gates $M_{\leq c}$ in the LTE, the Kolyvagin Euler system in BSD, the period matrix evaluation in the Hodge conjecture. Each gate carries irreducible LPO or CLASS cost.

Operation 3: Defer and localize. Restructure the proof so that the gates are invoked as late as possible and in as few places as possible. The algebraic skeleton should be complete before any gate is opened. This is what condensed mathematics does systematically via the sheaf-theoretic encoding. But it can also be done ad hoc: Paper 95's separation of Hecke arithmetic from Kolyvagin, Paper 80's separation of $\mathbb{Q}(t)$-algebra from analytic continuation, Paper 86's separation of Kani–Rosen splitting from Hodge-theoretic input.

The CRM series has been performing these operations informally for 96 papers. What condensed mathematics does—and what Scholze understood from the start—is systematize them at the categorical level. Our vocabulary (skeleton, gate, concentration ratio) is a rephrasing, not an advance.

The modest utility of CRMLINT here is quantitative triage: a proof where the scanner reports LPO in 2 of 15 modules is well-concentrated; a proof where LPO appears in 14 of 15 modules is diffuse. The diffuse case is where refactoring might help most. Whether CRMLINT actually helps anyone refactor remains to be demonstrated.

Observation C.4 (Concentration ratio)**.** Define the *concentration ratio* of a proof corpus C as

$$\rho(C) = 1 - \frac{\#\{\text{modules at LPO or above}\}}{\#\{\text{modules total}\}}.$$

For the LTE: $\rho = 1 - 10/15 = 0.33$. For a hypothetical "classical-diffuse" proof of the same theorem, $\rho = 0$. The concentration thesis predicts $\rho > 0$ for any proof that has undergone algebraic skeleton extraction. CRMLINT computes ρ automatically.

C.7 From Concentration to Excision: Predictive Applications

This part is prospective. The LTE analysis (§C.6) gave us a vocabulary: algebraic skeletons, analytic gates, and the strategy of *concentration*—deferring classical LPO or CLASS costs to isolated interface points.

However, the analysis of Scholze's Berkovich motives in the main body of this paper (Sections 3–4) reveals a vastly more powerful technique. By identifying the ℓ-adic comparison isomorphism $\iota : \overline{\mathbb{Q}_\ell} \cong \mathbb{C}$ as a *methodological* CLASS gate, Scholze bypassed it entirely by constructing a universal BISH skeleton: the motivic 6-functor formalism. We call this operation *motivic descent*. It does not merely concentrate the Archimedean obstruction; it physically excises it.

With motivic descent validated as a systemic bypass mechanism, we can transition the CRMLINT methodology from a retrospective diagnostic tool to a predictive radar. We now apply the Skeleton/Gate vocabulary to the open predictions of Section 8, providing an architectural blueprint for future formalization efforts.

C.7.1 The Weight-Monodromy Conjecture

Problem (Prediction 8.1). The weight-monodromy conjecture governs the action of the Weil group on the ℓ-adic cohomology of varieties over local fields. Classical proofs of known cases typically route through complex geometry (Betti realization, Hodge theory) to establish strict purity.

Concentration analysis.

(i) *Classical Gate:* The Betti–étale comparison isomorphism over $\mathbb{C}$ (cost: CLASS).

(ii) *Motivic Skeleton:* Rebuilding the weight spectral sequence natively within $D_{\mathrm{mot}}(X)$ using motivic weight structures.

Scanner deployment. A formalization blueprint attempting the motivic lift can be scanned by CRMLINT prior to coding. The scanner's role is to act as an Archimedean alarm system: if it detects markers like `Betti_realization` or `complex_conjugation`, the motivic lift has

leaked into the CLASS regime. If the scan peaks at WKL (flagging only the structural perfectoid/profinite inverse limits required for local field geometry), the methodology confirms the bypass is successful.

C.7.2 Function Field Langlands (BISH Floor)

Problem (Predictions 8.2 & 8.3). Lafforgue's global Langlands correspondence and Deligne's Ramanujan–Petersson bounds for $\mathrm{GL}_n/\mathbb{F}_q(C)$.

Concentration analysis. Over number fields, the global Arthur–Selberg trace formula is a *structural* CLASS gate, intrinsically requiring the L^2 spectral decomposition of the Archimedean factor $G(\mathbb{R})$. Over function fields $\mathbb{F}_q(C)$, the Archimedean place is physically absent (a domain transfer).

(i) *Motivic Skeleton:* The moduli stack of shtukas and its algebraic intersection theory natively in $DM(\mathbb{F}_q)$.

(ii) *Gate:* The Grothendieck–Lefschetz trace formula. Because this counts points on algebraic stacks over finite fields via ℓ-adic cohomology, its motivic lift is purely finitary algebraic geometry.

Scanner deployment. CRMLINT must be calibrated to strictly distinguish the algebraic Grothendieck–Lefschetz trace formula (BISH) from the analytic Arthur–Selberg trace formula (CLASS). A successful motivic blueprint for Lafforgue's theorem will scan as 100% BISH. Because discrete function fields require no analytic completions, even the WKL perfectoid cost is completely avoided.

C.7.3 The Shimura Boundary: A Positive Control

Problem (Prediction 8.5). Can the motivic descent strategy eliminate CLASS from the Langlands correspondence for Shimura varieties?

Concentration analysis. No. Shimura varieties are global objects defined over number fields; they inherently possess Archimedean places.

(i) *Skeleton:* The global adèle ring $\mathbb{A}_{\mathbb{Q}}$ and the geometry of the Shimura variety.

(ii) *Obstruction:* To match the motivic Galois representation to an automorphic form, the proof structurally requires Matsushima's formula and the global trace formula.

Scanner deployment. This provides a crucial *positive control* for the CRMLINT methodology. The CLASS gate here is *structural*, not *methodological*. If the scanner is applied to landmark global Langlands proofs (e.g., Harris–Taylor), it *must* light up with irreducible CLASS alerts: `Matsushima_formula`, `discrete_series`, `Plancherel_measure`, and `L2_cohomology`. This confirms that the scanner is not artificially blinding itself to classical logic; it strictly enforces the Archimedean Obstruction Thesis where the physics of the geometry demand it.

C.7.4 Legacy Upgrades: The BSD Conservation Gap

We can also turn this new motivic/analytic lens backward onto legacy problems from earlier in the CRM series, such as the Gross–Zagier–Kolyvagin proof of BSD (Papers 95–96).

Motivic strategy. Gate (1) in the BSD proof is the analytic behavior of the complex L-function near $s = 1$, a methodological CLASS gate. The motivic framework suggests an immediate upgrade path: bypass the complex L-function entirely by passing to a p-adic L-function (via overconvergent modular symbols). If the height formula can be evaluated using p-adic integration (which requires only WKL for inverse limits in the Iwasawa algebra) rather than complex integration, the CRM ceiling of the Gross–Zagier theorem drops structurally from CLASS to WKL. The scanner will cleanly track this bottleneck shift from `analytic_continuation` (CLASS) to `profinite_inverse_limit` (WKL).

C.7.5 The Updated Recipe: Motivic Proof Engineering

The discovery of Motivic Descent updates the proof-engineering loop from Section C.6. The general recipe for de-omniscientising a modern arithmetic geometry proof is now:

Step 1. Scan. Run CRMLINT on the initial LaTeX blueprint to map the CRM ceiling and concentration ratio ρ.

Step 2. Map the gates. Identify exactly which mathematical mechanisms trigger non-BISH alerts (the "bottleneck" field).

Step 3. Classify: Structural vs. Methodological. Determine if the gate is intrinsic to the theorem (like $M_{\leq c}$ in LTE, or $G(\mathbb{R})$ in Shimura varieties) or an artifact of the proof technique (like comparing ℓ-adic sheaves via $\mathbb{C}$).

Step 4. Attempt Motivic Descent. If the gate is methodological, attempt to lift the algebraic skeleton into a universal coefficient system (like D_{mot}) or execute a domain transfer *before* any realization functor is applied.

Step 5. Verify and Refactor. Re-scan the refactored blueprint. If a purely algebraic blueprint triggers a parasitic WLPO alert (e.g., an author carelessly writing an $\mathbb{R}_{\geq 0}$-valued norm instead of a decidable $\mathbb{Q}$-valued discrete group, as detected in the Berkovich audit), the team can refactor that definition to be discrete (BISH) *before* writing a single line of Lean 4 code.

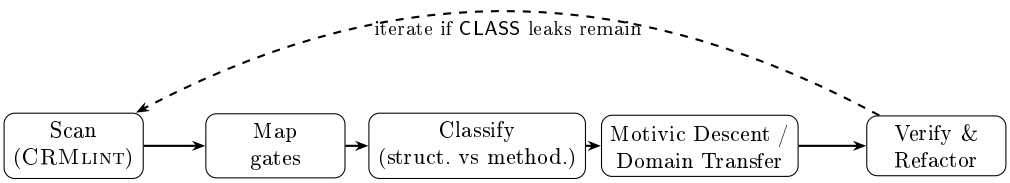


Figure 1: The upgraded scan-map-classify-search-verify loop, featuring Motivic Descent.

The ultimate promise of this methodology is economic. By running CRMLINT on a LaTeX blueprint, a formalization team can identify and mathematically excise classical parasitic dependencies, saving thousands of hours of formalization labor by ensuring the architecture is constructively optimal before the programming begins.

C.8 The Mathematics of the LTE

This part is expository. We restate the main theorems of the LTE in the form proved by Scholze, Commelin, and Massot, and annotate each step with its CRM level. Nothing here is new mathematics. The proofs are theirs; the CRM labels are ours. The labels may be useful for readers approaching the LTE from the constructive side, or for formalization teams deciding which components to port to Lean 4.

C.8.1 The main theorem and its reduction

The LTE establishes the following vanishing result, which Scholze proposed as a challenge in December 2020.

Theorem C.5 (Clausen–Scholze [22]). *Let $0 < p' < p \leq 1$ be real numbers, let S be a profinite set, and let V be a p-Banach space. Let $\mathcal{M}_{p'}(S)$ denote the space of real p'-measures on S. Then*

$$\mathrm{Ext}^i_{\mathrm{Cond(Ab)}}(\mathcal{M}_{p'}(S), V) = 0 \qquad \textit{for all } i \geq 1.$$

The Ext-groups are computed in the abelian category of condensed abelian groups. The theorem says: from the perspective of homological algebra in the condensed world, real-valued measures are "projective-like" — they have no higher Ext against Banach targets.

CRM annotation. The *statement* of Theorem C.5 requires:

- p-Banach space V: the norm $\|\cdot\|$ on V satisfies $\|x+y\|^p \leq \|x\|^p + \|y\|^p$. Defining this norm and its completeness is WLPO.
- The condensed abelian group $\mathcal{M}_{p'}(S)$: defined via a filtration $\mathcal{M}_{p'}(S)_{\leq c}$, each level a compact Hausdorff space. The filtration cut-off is LPO.
- Profinite S: a cofiltered limit of finite sets. All BISH.

The *proof* reduces Theorem C.5 to a chain of intermediate results, each of which we now state.

C.8.2 The proof chain: six links

The proof proceeds by six reductions, from the main theorem down to the key technical result (Theorem C.9). We trace the full chain and label each link with its CRM level.

Link	Result	Mechanism	CRM
6	Theorem C.5 (main)	Long exact Ext sequence from Link 5	BISH
5	Ext-vanishing for $\mathcal{L}_{r'}(S)$	T^{-1}-bijectivity via Link 4	BISH
4	Ext-vanishing for $\overline{\mathcal{L}}_{r'}(S)$	Pro-isomorphism + derived limit via Link 3	BISH
3	Theorem C.9 (Ext-vanishing at finite level)	Double complex + spectral proposition	LPO
2	Column exactness (Proposition C.8)	Čech nerve + normed snake lemma	LPO
1	Čech exactness (Proposition C.6)	Contracting homotopy + profinite limit	BISH/WKL

Link 1: Čech nerve exactness (BISH/WKL)

Proposition C.6 (Čech cover exactness [23]). *Let $\pi : X \to B$ be a surjective morphism of profinite sets, and let $S_\bullet \to S_{-1}$, $S_{-1} := B$, be its augmented Čech nerve. Let V be a semi-normed group. Then the complex*

$$0 \longrightarrow \widehat{V}(S_{-1}) \longrightarrow \widehat{V}(S_0) \longrightarrow \widehat{V}(S_1) \longrightarrow \cdots$$

is exact. Moreover, for every $\varepsilon > 0$ and $f \in \ker(\widehat{V}(S_m) \to \widehat{V}(S_{m+1}))$, there exists $g \in \widehat{V}(S_{m-1})$ with $d(g) = f$ and $\|g\| \leq (1+\varepsilon)\|f\|$.

Proof sketch. For *finite* X and B, a section $\sigma : B \to X$ of π induces maps $S_m \to S_{m+1}$ via $(a_0, \ldots, a_m) \mapsto (\sigma(\pi(a_0)), a_0, \ldots, a_m)$. These provide a contracting homotopy for the complex, and the homotopy maps are norm-nonincreasing. For general profinite sets, write $X = \varprojlim_i X_i$ over its discrete (finite) quotients, pass to finite quotients $X_i \to B_i$, apply the finite case, and take the cofiltered limit. Surjectivity of transition maps between compact Hausdorff spaces is preserved under cofiltered limits (by Tychonoff).

CRM annotation. The finite case is entirely BISH: the section σ exists because π is surjective between finite sets, and the contracting homotopy is an explicit combinatorial construction. The passage to profinite limits uses Tychonoff's theorem for compact Hausdorff spaces (surjections of compact spaces remain surjective under cofiltered limits). In CRM terms, Tychonoff for Hausdorff spaces is WKL (equivalent to weak König's lemma: an infinite finitely-branching tree has an infinite path). The norm estimates ($\|g\| \leq (1+\varepsilon)\|f\|$) do not introduce additional logical cost.

Link 2: Column exactness via normed snake lemma (LPO) The double complex construction requires that each column is "weakly bounded exact." This is established by combining Čech exactness (Link 1) with the normed T^{-1}-action.

Definition C.7 (Weak bounded exactness [23]). A system of complexes $C^\bullet_\bullet$ is *weakly $\leq k$-exact in degrees $\leq m$ for $c \geq c_0$ with bound K* if for all $c \geq c_0$, all $x \in C^i_{kc}$ with $i \leq m$, and any $\varepsilon > 0$, there exists $y \in C^{i-1}_c$ with

$$\|x_{|c} - dy\| \leq K\|dx\| + \varepsilon.$$

Proposition C.8 (Column exactness [23]). *Let $k > 1$, $c_0 > 0$ with $(k-1)c_0/N \geq d$ (the splittability error). Let m be any natural number, and put*

$$K = (m+2) + \frac{r+1}{r(1-r)}(m+2)^2.$$

Then the i-th column in the double complex is (k^2, K)-weakly bounded exact in degrees $\leq m$ for $c \geq c_0'$.

Proof sketch. The key input is the "normed dual snake lemma" (Proposition C.12 below): given two admissible systems of complexes $M^\bullet_\bullet$ and $M'^\bullet_\bullet$ with a map $f : M \to M'$, if both M and M' are weakly bounded exact, then the kernel complex inherits weak bounded exactness (with controlled loss in the constants). The column is built from the Čech nerve of the summation map $M^N_{c/N} \to M_c$, and the T^{-1}-action introduces the norm scaling.

CRM annotation. The splittability lemma (N-splittability of $\mathrm{Hom}(\Lambda, \overline{\mathcal{L}}_{r'}(S))$) is BISH: it decomposes elements of a polyhedral lattice module into N pieces of roughly $1/N$-th the norm. The column exactness proof combines this with the LPO content of the filtration cut-offs $M_{\leq c}$. The constant K depends on m and r (not on the profinite set S or the normed module V), so the "universal" part of the proof is BISH and the LPO enters only at the interface where elements are tested against filtration levels.

Link 3: The key technical result (LPO)

Theorem C.9 (Theorem 9.4 [21, 23]). *Let $\mathrm{BD} = (n, f, h)$ be a Breen–Deligne package, and let $\kappa = (\kappa_0, \kappa_1, \ldots)$ be a sequence of constants in $\mathbb{R}_{\geq 0}$ that is BD-suitable. Fix radii $1 > r' > r > 0$. For any m there exist k and c_0 such that for all profinite sets S and all r-normed $\mathbb{Z}[T^{\pm 1}]$-modules V, the system of complexes*

$$C^{\mathrm{BD}}_\kappa(\overline{\mathcal{L}}_{r'}(S))^\bullet_c\colon\ \widehat{V}(\overline{\mathcal{L}}_{r'}(S)_{\leq c})^{T^{-1}} \longrightarrow \widehat{V}(\overline{\mathcal{L}}_{r'}(S)^2_{\leq \kappa_1 c})^{T^{-1}} \longrightarrow \cdots$$

is $\leq k$-exact in degrees $\leq m$ for $c \geq c_0$.

Proof structure. The proof is by induction on m. The inductive step requires a *stronger* statement (Theorem 9.5′ in the blueprint), which introduces a polyhedral lattice Λ and replaces $\overline{\mathcal{L}}_{r'}(S)$ by $\mathrm{Hom}(\Lambda, \overline{\mathcal{L}}_{r'}(S))$. Theorem C.9 is recovered by setting $\Lambda = \mathbb{Z}$.

The inductive step constructs a *system of double complexes*:

1. **Rows:** Apply the Breen–Deligne system functor to $\mathrm{Hom}(\Lambda^{(\bullet)}, M)$, where $\Lambda^{(\bullet)}$ is the cosimplicial polyhedral lattice and M abbreviates $\overline{\mathcal{L}}_{r'}(S)$. Take the alternating face map complex. Rescale norms by $m!$ for admissibility.
2. **Columns:** The column at index i is the Čech nerve construction of Link 2, applied with the i-th Breen–Deligne data.
3. **Homotopy:** The Breen–Deligne homotopy h^N (the N-fold composition of the package's built-in homotopy) provides maps $M^{0,q+1}_{k'c} \to M^{1,q}_c$ bounded by some constant H. The "homotopic map" $\delta^{0,q}_c$ factors through a deep restriction and has norm $\leq \varepsilon$.

The three ingredients (row exactness by induction, column exactness from Link 2, homotopy condition) are fed into the *spectral proposition* (Proposition C.13 below), which yields weak bounded exactness of the first row—completing the induction.

CRM annotation. The double complex construction (cosimplicial lattice, alternating face map, norm rescaling) is BISH: it is combinatorial algebra over $\mathbb{Z}$. The Breen–Deligne resolution itself is BISH: it is a functorial resolution of abelian groups by free modules, specified by integer matrices. The homotopy bound H is computed from the Breen–Deligne data and r, r'—a finite computation over $\mathbb{Q}$, hence BISH.

The LPO enters at two points: (i) the filtration cut-offs $\overline{\mathcal{L}}_{r'}(S)_{\leq c}$ in the system of complexes, and (ii) the norm estimate $r^b N \leq \varepsilon$ (choosing b so that $2k'(r/r')^b \leq \varepsilon$), which requires comparing real numbers. The choice of N as a power of 2 satisfying $k'/N \leq (r')^b$ is a single LPO decision on the ordered reals.

Links 4–6: Algebraic reductions (BISH) Link 4: From finite level to $\overline{\mathcal{L}}_{r'}(S)$. Write $Q'(\overline{\mathcal{M}}_{r'}(S)) = \varinjlim_c Q'(\overline{\mathcal{M}}_{r'}(S))_{\leq c}$ and reduce to a pro-isomorphism statement at each finite level. The passage to the derived limit uses a resolution $0 \to \bigoplus_n Q'_{\leq n} \to \bigoplus_n Q'_{\leq n} \to Q' \to 0$ and checks that the relevant squares are bicartesian (vertical kernels and cokernels form pro-zero systems). This is diagram chasing in Cond(Ab), entirely BISH.

Link 5: From $\overline{\mathcal{L}}_{r'}(S)$ to $\mathcal{L}_{r'}(S)$. The short exact sequence

$$0 \longrightarrow \mathbb{Z}[T^{-1}][S] \longrightarrow \mathcal{L}_{r'}(S) \longrightarrow \overline{\mathcal{L}}_{r'}(S) \longrightarrow 0$$

decomposes Laurent measures into negative and positive parts. Apply $\mathrm{Ext}^*(_, V)$: the $\mathbb{Z}[T^{-1}][S]$-term has vanishing higher Ext (because $\mathbb{Z}[S]$ is free), and the $\overline{\mathcal{L}}_{r'}(S)$-term vanishes by Link 4. The five lemma gives vanishing for $\mathcal{L}_{r'}(S)$. This is abstract homological algebra, all BISH.

Link 6: Main theorem. The short exact sequence

$$0 \longrightarrow \mathcal{L}_{r'}(S) \xrightarrow{2T-1} \mathcal{L}_{r'}(S) \longrightarrow \mathcal{M}_{p'}(S) \longrightarrow 0$$

(evaluation at $T = 1/2$) reduces the main theorem to the Ext-vanishing of $\mathcal{L}_{r'}(S)$ from Link 5 via the long exact sequence. The surjectivity of evaluation at $T = 1/2$ on pseudo-normed groups is a norm estimate (strong exactness), which uses Tychonoff (WKL) for passage to profinite limits. But the long-exact-sequence argument itself is BISH.

C.8.3 The normed homological algebra toolkit

The LTE's main technical innovation is not the theorem statement but the *toolkit*: a library of normed homological algebra results with explicit quantitative control. These results are general-purpose and transfer directly to other formalization targets.

Controlled surjectivity

Lemma C.10 (Closure surjectivity [23]). *Let G and H be normed groups, $K \leq H$ a subgroup, and $f : G \to H$ a C-surjective morphism onto K (i.e., for all $x \in K$ there exists g with $f(g) = x$ and $\|g\| \leq C\|x\|$). If G is complete, then f is $(C + \varepsilon)$-surjective onto $\overline{K}$ for every $\varepsilon > 0$.*

Proof idea. Write $x \in \overline{K}$ as $x = \sum_{i \geq 0} x_i$ with $x_i \in K$ and $\|x_i\| \leq \varepsilon_i$ for a rapidly decreasing sequence. Lift each x_i to g_i with $\|g_i\| \leq C\|x_i\|$, and set $g = \sum g_i$. Completeness of G guarantees convergence. The norm estimate $\|g\| \leq (C + \varepsilon)\|x\|$ follows from the geometric series bound on the ε_i.

CRM annotation. The proof uses completeness of G (to sum the Cauchy series $\sum g_i$), which is WLPO (Cauchy convergence test). The lifting $x_i \mapsto g_i$ uses the C-surjectivity hypothesis, which is BISH (explicit choice of lifts). The norm comparison $\|g_i\| \leq C\|x_i\|$ is an inequality between real numbers (LPO).

Controlled exactness

Lemma C.11 (Controlled exactness [23]). *Let $f : M_0 \to M_1$ and $g : M_1 \to M_2$ be bounded maps between normed groups. Suppose f is C-surjective onto $\ker g$, and g is D-surjective onto its image. Then for every $\varepsilon > 0$, the completion $\widehat{f}$ is $(C+\varepsilon)$-surjective onto $\ker \widehat{g}$.*

This is the mechanism by which the LTE passes from "incomplete" exactness (at the pre-completion level, BISH) to "completed" exactness (at the normed-completion level, WLPO). The upgrade is Lemma C.10 applied to $K = \ker g$ inside $\widehat{M_1}$.

The normed snake lemma

Proposition C.12 (Weak normed snake lemma [23]). *Let $M^\bullet_\bullet$ and $M'^\bullet_\bullet$ be two admissible systems of complexes of complete normed abelian groups, with a norm-nonincreasing map $f : M \to M'$ satisfying $\|x_{|c}\| \le K''\|f(x)\|$ for all $x \in M^i_{k''c}$, $i \le m+1$. If M is weakly $\le k$-exact with bound K and M' is weakly $\le k'$-exact with bound K', then the quotient $N = M'/M$ is weakly $\le kk'k''$-exact in degrees $\le m-1$ with bound $K'(KK''+1)$.*

CRM annotation. The proof is a quantitative diagram chase: lift $n \in N^i$ to $m' \in M'^i$, decompose dm' using the quotient norm, apply weak exactness of M and M', and track constants through. The diagram chase is BISH; the norm estimates are LPO. The bound $K'(KK''+1)$ is explicit and computable.

The *dual* snake lemma (for kernels instead of cokernels) is proved analogously, with bound $K + r_1 r_2 K K'$. Both versions are used in the inductive step of Theorem C.9: the snake lemma produces column exactness, and the dual snake lemma produces the kernel complex for the T^{-1}-action.

The spectral proposition

Proposition C.13 (Spectral proposition [23]). *Fix $m \ge 0$ and constants k, K. There exist $\varepsilon > 0$ and k_0, K_0 (depending only on k, K, m) such that: if a system of double complexes $M^{p,q}_c$ is admissible and satisfies*

(i) rows are weakly $\le k$-exact in degrees $\le m-1$ with bound K (for $i = 0, \dots, m+1$),
(ii) columns are weakly $\le k$-exact in degrees $\le m$ with bound K,
(iii) the normed spectral homotopy condition for m, some $H > 0$, c_0, and ε,

then the first row is weakly $\le k'^2$-exact in degrees $\le m$ with bound $2K_0H$.

This is the engine of the induction. The base case ($m = 0$) uses $\varepsilon = 1/(2k)$ and the homotopy condition to absorb the ε-term. The inductive step passes to the quotient complex $N^{p,0} = M^{p,1}/\overline{M^{p,0}}$ and applies the snake lemma (Proposition C.12) to verify the hypotheses for $m-1$.

CRM annotation. The spectral proposition is BISH in its logical content: it is a statement about constants and admissibility conditions. The LPO enters only when the proposition is *applied* to a specific system of double complexes built from normed groups. This separation is a paradigm case of the concentration thesis: the abstract machinery is logically free, and the analytic cost is paid only at the point of application.

C.8.4 The Breen–Deligne resolution: pure algebra

The deepest algebraic ingredient in the LTE is the Breen–Deligne resolution, which provides a functorial free resolution of any abelian group A.

Theorem C.14 (Breen–Deligne [36]). *For any abelian group A, there is a resolution, functorial in A:*

$$\cdots \longrightarrow \bigoplus_i \mathbb{Z}[A^{r_{ij}}] \longrightarrow \cdots \longrightarrow \mathbb{Z}[A^3] \oplus \mathbb{Z}[A^2] \longrightarrow \mathbb{Z}[A^2] \longrightarrow \mathbb{Z}[A] \longrightarrow A \longrightarrow 0.$$

The functorial maps $\mathbb{Z}[A^m] \to \mathbb{Z}[A^n]$ are specified by integer matrices ("basic universal maps": $n \times m$ matrices over $\mathbb{Z}$), composed via matrix multiplication.

CRM annotation. The entire Breen–Deligne resolution—its existence, functoriality, and the universal maps—is BISH. The universal maps are $\mathbb{Z}$-linear combinations of basic universal maps specified by integer matrices; their composition is matrix multiplication; the homotopy h packaged with the resolution is a universal map bounded by an integer N. No real analysis, no completeness, no decidability of real arithmetic appears anywhere. Just as Scholze's motivic geometric Satake achieves BISH by replacing analytic intersection cohomology with algebraic MV cycles (§3), the Breen–Deligne resolution achieves BISH by replacing analytic bounds with integer matrices.

This is the algebraic skeleton of the LTE in its purest form. The LTE formalizes it in Lean 3 as `breen_deligne.basic_universal_map` (integer matrices), `breen_deligne.universal_map` (formal $\mathbb{Z}$-linear combinations), and `breen_deligne.package` (the triple (n, f, h)).

C.8.5 What transfers: reusable components

The LTE's mathematical infrastructure divides into three categories by reusability.

Category A: Directly reusable (BISH) These components depend on no analytic input and can be ported to Lean 4 as standalone libraries.

1. **Breen–Deligne package.** Integer matrices, universal maps, the package triple (n, f, h), homotopy composition h^N, suitability conditions on κ. All finitary algebra over $\mathbb{Z}$.
2. **Polyhedral lattices.** Finite free abelian groups with generating norms. The key combinatorial lemma: N-splittability of $\mathrm{Hom}(\Lambda, \overline{\mathcal{L}}_{r'}(S))$ with explicit error term d.
3. **Cosimplicial polyhedral lattices.** $\Lambda' = \Lambda^N$ with the ℓ^1-norm; the cosimplicial object $\Lambda^{(\bullet)}$; the Čech nerve isomorphisms $\mathrm{Hom}(\Lambda'^{(m)}, M)_{\leq c} \cong \mathrm{Hom}(\Lambda', M)_{\leq c}^{m/\mathrm{Hom}(\Lambda, M)_{\leq c}}$.
4. **Abstract diagram chasing.** The pro-isomorphism argument (Link 4), the five-lemma reduction (Link 5), the short-exact-sequence reduction (Link 6).

Category B: Reusable with LPO interface (LPO) These components are algebraic in structure but require norm comparisons at the interface.

1. **Systems of complexes.** Functors $(\mathbb{R}_{\geq 0})^{\mathrm{op}} \to \mathrm{Ch}(\mathrm{SemiNormGrp})$. The restriction maps $\mathrm{res}_{c',c}$ require $c' \geq c$ (LPO). The definitions of $\leq k$-exactness, weak $\leq k$-exactness, and admissibility are BISH conditions on the norms.
2. **The spectral proposition.** The statement and proof of Proposition C.13 are BISH as abstract algebra. The LPO enters only at instantiation.
3. **The normed snake lemma (both versions).** The diagram chase is BISH; the norm bounds use LPO.

Category C: Analytic core (WLPO/LPO) These components are irreducibly analytic.

1. **Normed completions $\widehat{V}(\cdot)$.** The completion functor on semi-normed groups. Cauchy convergence is WLPO; the norm on the completion is LPO.
2. **The T^{-1}-action on $\overline{\mathcal{L}}_{r'}(S)$.** The shift $T^{-1} \cdot (\sum a_n T^n) = (\sum a_{n+1} T^n)$ scales the norm by r': the filtration level changes from $\leq c$ to $\leq c/r'$. This scaling requires norm comparison (LPO).
3. **The homotopy norm estimate.** The key bound $r^b N \leq \varepsilon$ (choosing b so that $2k'(r/r')^b \leq \varepsilon$) requires decidable ordering on $\mathbb{R}$ (LPO). This is the deepest analytic content in the entire proof: a single real-number comparison that controls the convergence of the induction.

C.8.6 Lean 4 formalization targets

The LTE was formalized in Lean 3 (community `mathlib` fork). A Lean 4 port faces the standard Lean 3 → Lean 4 translation burden, but the *mathematical* targets are well-defined. We identify the most valuable targets from the CRM perspective: components where the BISH/LPO separation is sharpest and the reuse potential is highest.

Target 1: Breen–Deligne package (pure BISH)

```
-- Basic universal maps: integer matrices
structure BasicUniversalMap (m n : Nat) where
  matrix : Matrix (Fin n) (Fin m) Int

-- Universal maps: formal Z-linear combinations
structure UniversalMap (m n : Nat) where
  terms : List (Int x BasicUniversalMap m n)

-- Composition via matrix multiplication
def UniversalMap.comp (f : UniversalMap m n)
    (g : UniversalMap l m) : UniversalMap l n := ...

-- BD package: exponents, differentials, homotopy
structure BDPackage where
  n : Nat -> Nat                  -- exponents
  f : (i : Nat) -> UniversalMap (n (i+1)) (n i)
  h : (i : Nat) -> UniversalMap (n (i+1)) (n i)
  chain_complex : forall i, (f i).comp (f (i+1)) = 0
  homotopy_rel : forall i, ...   -- h witnesses null-homotopy
```

This is finite combinatorics over $\mathbb{Z}$. No axioms beyond `propext` and `Quot.sound`. The key theorem to formalize: given a BD package and a suitable sequence κ, the N-fold homotopy h^N exists and is bounded by an explicit integer.

Target 2: Systems of complexes (BISH definitions, LPO at interface)

```
-- System of complexes: parametrized by NNReal
structure SystemOfComplexes where
  C : NNReal -> ChainComplex SemiNormedGroup
  res : forall c' c, c' >= c -> C c' --> C c
  res_comp : forall c'' c' c (h1 : c'' >= c')
    (h2 : c' >= c), res c'' c (le_trans h2 h1) =
    (res c' c h2).comp (res c'' c' h1)

-- Weak bounded exactness
def SystemOfComplexes.IsWeakBoundedExact
    (S : SystemOfComplexes) (k K : Real) (m : Nat)
    (c0 : NNReal) : Prop :=
  forall c >= c0, forall i <= m,
    forall x in (S.C (k * c)).obj i,
    forall eps > 0, exists y in (S.C c).obj (i-1),
      norm (res x - d y) <= K * norm (d x) + eps
```

The definition is BISH. The LPO enters at the inequality $c' \geq c$ (comparison of real numbers) and the norm estimate.

Target 3: Controlled surjectivity lemma (WLPO)

```
-- Key lemma: C-surjectivity extends to closures
theorem controlled_closure_of_complete
    {G H : NormedAddCommGroup} [CompleteSpace G]
    {K : AddSubgroup H} {f : G ->+ H} {C : Real}
    (hf : forall x in K,
      exists g, f g = x && norm g <= C * norm x)
    (eps : Real) (heps : 0 < eps) :
    forall x in K.topologicalClosure,
      exists g, f g = x
        && norm g <= (C + eps) * norm x := by
  -- Proof: geometric series in complete G
  ...
```

This requires `CompleteSpace G` (WLPO: Cauchy sequences converge) and the norm bound (LPO: $\|g\| \leq C\|x\|$). The Lean 4 proof would use `Metric.cauchySeq_of_le_geometric` from Mathlib.

Target 4: Normed snake lemma (BISH chase, LPO bounds) The most technically demanding target. The Lean 3 proof (`weak_normed_snake` and `weak_normed_snake_dual`) runs approximately 200 lines. A Lean 4 port would:

- Define admissible systems of complexes over Mathlib's `SemiNormedGroup`.
- Implement the quantitative diagram chase with explicit ε-management.
- Prove the bound $K'(KK''+1)$ for quotients and $K + r_1 r_2 K K'$ for kernels.

The entire chase is BISH (diagram manipulation in an abelian category with explicit lifts). The LPO enters only at the final norm inequality.

CRM summary of formalization targets

Target	Component	CRM	Lean 4 difficulty
1	Breen–Deligne package	BISH	Low (finite algebra over $\mathbb{Z}$)
2	Systems of complexes	BISH/LPO	Medium (parametrized complexes)
3	Controlled surjectivity	WLPO	Low (standard Cauchy argument)
4	Normed snake lemma	BISH/LPO	High (200-line diagram chase)

Targets 1 and 3 are low-hanging fruit: they are self-contained, well-typed, and already have Lean 3 implementations to translate. Target 2 requires design decisions about how to interface with Mathlib's `CategoryTheory.ChainComplex`. Target 4 is the highest-value formalization: the normed snake lemma is the engine of the induction, and a clean Lean 4 version would serve as infrastructure for any future normed homological algebra project.

Acknowledgments for this appendix. The LTE blueprint files used in this analysis were authored by Johan Commelin and Patrick Massot [23]. The mathematical content follows the proofs in [21] as formalized in [23].

References

[1] P. Scholze, Berkovich motives, arXiv:2412.03382, 2024 (revised January 2026).

[2] P. Scholze, Geometrization of the local Langlands correspondence, motivically, arXiv:2501.07944, January 2026.

[3] L. Fargues and P. Scholze, Geometrization of the local Langlands correspondence, arXiv:2102.13459, to appear in Astérisque, 2021.

[4] P. C.-K. Lee, The motivic CRM architecture: why the six domains of mathematics have different logical signatures, and how motives explain the pattern (Paper 98), Zenodo, 2026. DOI: 10.5281/zenodo.18828345.

[5] T. Richarz and J. Scholbach, The motivic Satake equivalence, *Math. Ann.* 380 (2021), 1595–1653.

[6] R. Cass, T. van den Hove, and J. Scholbach, The geometric Satake equivalence for integral motives, preprint.

[7] E. Bishop, *Foundations of Constructive Analysis*, McGraw-Hill, 1967.

[8] D. Bridges and F. Richman, *Varieties of Constructive Mathematics*, LMS Lecture Notes 97, Cambridge, 1987.

[9] H. Ishihara, Reverse mathematics in Bishop's constructive mathematics, *Phil. Sci. Cahier Spécial* 6 (2006), 43–59.

[10] A. Wiles, Modular elliptic curves and Fermat's Last Theorem, *Ann. of Math.* 141 (1995), 443–551.

[11] R. Taylor and A. Wiles, Ring-theoretic properties of certain Hecke algebras, *Ann. of Math.* 141 (1995), 553–572.

[12] L. Lafforgue, Chtoucas de Drinfeld et correspondance de Langlands, *Invent. Math.* 147 (2002), 1–241.

[13] J. Arthur, An introduction to the trace formula, in: *Harmonic Analysis, the Trace Formula, and Shimura Varieties*, Clay Math. Proc. 4, AMS, 2005, 1–263.

[14] M. Hochster, Prime ideal structure in commutative rings, *Trans. Amer. Math. Soc.* 142 (1969), 43–60.

[15] P. C.-K. Lee, CRMlint: automated logical cost analysis of mathematical proof (Paper 76), Zenodo, 2026. DOI: 10.5281/zenodo.18779362.

[16] P. C.-K. Lee, The Hecke theta series squeeze: resolving the dihedral base case of modularity via Mumford's algebraic theta functions (Paper 99), Zenodo, 2026.

[17] P. C.-K. Lee, The Kuga–Satake bifurcation: absolute Hodge classes on K3 surfaces via Shioda–Inose structure (Paper 100), Zenodo, 2026.

[18] P. C.-K. Lee, CRMlint scanner output and classification for Berkovich motives and motivic geometrization, supplementary appendix to Paper 101, 2026.

[19] P. C.-K. Lee, Paper format guide for the constructive reverse mathematics series, Zenodo, 2026. DOI: 10.5281/zenodo.18765700.

[20] K. Buzzard, J. Commelin, and P. Massot, Formalising perfectoid spaces, in: *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, 2020, 299–312.

[21] P. Scholze, Lectures on analytic geometry, lecture notes, 2019.

[22] P. Scholze, Liquid tensor experiment, 2020. Available at `https://xenaproject.wordpress.com/2020/12/05/`.

[23] J. Commelin and P. Massot, Liquid tensor experiment: formal verification in Lean 3, 2022. Repository: https://github.com/leanprover-community/liquid.

[24] D. Clausen and P. Scholze, *Condensed Mathematics and Complex Geometry*, 2022.

[25] P. C.-K. Lee, General relativity is BISH + LPO: the logical cost of the equivalence principle (Paper 5), CRM Series, 2024. DOI: 10.5281/zenodo.18765700.

[26] P. C.-K. Lee, The DPT characterisation theorem (Paper 72), CRM Series, 2025.

[27] P. C.-K. Lee, Axiom 1 is irreversible (Paper 73), CRM Series, 2025.

[28] P. C.-K. Lee, Axiom 2 is irreversible (Paper 74), CRM Series, 2025.

[29] P. C.-K. Lee, Explicit Hodge decompositions for E^4 (Paper 77), CRM Series, 2026. DOI: 10.5281/zenodo.18777596.

[30] P. C.-K. Lee, The local Langlands correspondence for $\mathrm{GL}_2(\mathbb{Q}_2)$ is constructively decidable (Paper 78), CRM Series, 2026. DOI: 10.5281/zenodo.18789008.

[31] P. C.-K. Lee, The squeeze is a microscope: automated constructivization from CRMLint to the Hodge horizon (Paper 90), CRM Series, 2026. DOI: 10.5281/zenodo.18809911.

[32] P. C.-K. Lee, The BSD squeeze: CRMLint decomposition of the Gross–Zagier–Kolyvagin proof (Paper 95), CRM Series, 2026. DOI: 10.5281/zenodo.18821019.

[33] P. C.-K. Lee, The root number bifurcation: CRM cost of BSD detection controlled by the functional equation (Paper 96), CRM Series, 2026. DOI: 10.5281/zenodo.18820969.

[34] The Mathlib Community, *Mathlib4: Mathematics in Lean 4*, https://github.com/leanprover-community/mathlib4, 2024–2026.

[35] S. G. Simpson, *Subsystems of Second Order Arithmetic*, 2nd ed., Cambridge University Press, 2009.

[36] L. Breen, Extensions du groupe additif, *Inst. Hautes Études Sci. Publ. Math.* 48 (1978), 39–125.

[37] A. N. Tychonoff, Über die topologische Erweiterung von Räumen, *Math. Ann.* 102 (1930), 544–561.